



COSMETIC PRODUCT SAFETY REPORT

For:

DERMOTIN BIOWART

In accordance with Regulation (EC) 1223/2009 of the European Parliament and of the Council of 30 November 2009 on cosmetic products

Responsible Person: Za Republiku Srbiju: Clicky doo, Zvezdarskih Jelki 3/2, Beograd

EU: Clicky EOOD, Traditsa 6, Sofia, Bulgaria

Date: 20.11.2024.

Document version: 1

Safety Assessor: Vesna Milanković

Part A. Cosmetic Product Safety Information

Trade name: DERMOTIN BIOWART

Volume: 50ml

Product reference number or barcode:

Product category: Skin emulsion

Manufacturer: BIO-TEO centar d.o.o., Mitropolita Mraovića 1a, 11000 Belgrade, Serbia

1) Qualitative and quantitative composition of the product

INCI	Concentration of raw material in the product (%)	Raw material composition (%)	CAS	EINECS/ELINCS	Function	Impurities
STEARIC ACID	1.97	100	57-11-4	200-313-4	emolijens	N/A
GLYCERYL STEARATE SE	1.28	100	11099-07-3	234-325-6	emulgator	N/A
CETYL ALCOHOL	1.46	100	36653-82-4	253-149-0	emolijens	N/A
SYNTHETIC BEESWAX	1.10	100	71243-51-1	275-286-5	emolijens	N/A
PHENOXYETHANOL	0.91	75-100	122-99-6	204-589-7	konzervans	N/A
ETHYLHEXYLGLYCERIN		10-25	70445-33-9	408-080-2	stabilizator	
PRUNUS AMYGDALUS DULCIS OIL	0.91	100	8007-69-0 / 90320-37-9	- / 291-063-5	emolijens	N/A
AQUA	61.20	100	7732-18-5	231-791-2	vodena faza	N/A
CARBOMER	0.27	100	9007-20-9 / 9003-01-4 / 76050-42-5 / 9062-04-8 / 9007-16-3 / 9007-17-4	- / - / - / - / - / -	reoloski modifikator	Ethyl Acetate 0.23%, Cyclohexane 0.02%
GLYCERIN	1.83	100	56-81-5	200-289-5	humektans	N/A
PANTHENOL	0.91	100	81-13-0 / 16485-10-2	201-327-3 / 240-540-6	kas	N/A
UREA	0.91	100	57-13-6	200-315-5	kas	N/A
POTASSIUM SORBATE	0.09	100	24634-61-5 / 590-00-1	246-376-1 / -	konzervans	Formaldehyde 0.1%
SODIUM BENZOATE	0.18	100	532-32-1	208-534-8	konzervans	N/A
TRIETHANOLAMINE	0.75	100	102-71-6	203-049-8	podesavanje pH	N/A
CALENDULA OFFICINALIS FLOWER EXTRACT	25.58	7-8.5	84776-23-8	283-949-5	kas	N/A

CHELIDONIUM MAJUS EXTRACT		7-8.5	84603-56-5	283-257-3	kas	
ACHILLEA MILLEFOLIUM EXTRACT		3-4.5	84082-83-7	282-030-6	kas	
GALIUM VERUM EXTRACT		3-4.5	90028-84-5	289-878-6	kas	
HYPERICUM PERFORATUM EXTRACT		3-4.5	84082-80-4	282-026-4	kas	
QUERCUS ROBUR BARK EXTRACT		3-4.5	71011-28-4	275-129-0	kas	
SALVIA OFFICINALIS LEAF EXTRACT		3-4.5	84082-79-1 / 8022-56-8	282-025-9 / -	kas	
ALCOHOL		20-40	64-17-5	200-578-6	rastvarac	
AQUA		20-40	7732-18-5	231-791-2	rastvarac	
TOCOPHERYL ACETATE	0.09	100	7695-91-2 / 58-95-7	231-710-0 / 200-405-4	antioksidans	N/A
RETINYL PALMITATE	0.09	100	79-81-2	201-228-5	kas	N/A
SALVIA OFFICINALIS OIL	0.06	100	8022-56-8 / 84776-73-8	- / 282-025-9	kas	Limonene <5%
EUCALYPTUS GLOBULUS LEAF OIL	0.06	100	8000-48-4 / 84625-32-1	- / 283-406-2	kas	Limonene <15%
MENTHA PIPERITA OIL	0.06	100	8006-90-4 / 84082-70-2	- / 282-015-4	kas	Limonene <1-3.5%
EUGENIA CARYOPHYLLUS BUD OIL	0.06	100	84961-50-2	284-638-7	kas	Eugenol 75-88%
THYMUS VULGARIS FLOWER/LEAF OIL	0.06	100	84929-51-1 / 8007-46-3	284-535-7 / -	kas	Linalool 1.5-6.5%
ILICIIUM VERUM LEAF OIL	0.06	100	84650-59-9	283-518-1 (I)	kas	Limonene <5%, Linalool <2%
CITRUS LIMON PEEL OIL	0.06	100	8008-56-8 / 84929-31-7	- / 284-515-8	kas	Citral <3.5%, Limonene <70%

*Limonene, Linalool, Eugenol and Citral must be labelled

2) Physical and chemical characteristics and stability of the cosmetic product

a) Physical and chemical characteristics of the raw materials

For all raw materials used for production of this cosmetic product technical documentation (SDS, TDS, CoA, etc.) provided by the supplier of cosmetic ingredients is available and is attached to this document. After evaluating the documents provided, it was concluded that all the ingredients used for production of cosmetic product GINGERIA GROW HAIR CARE met the requirements of quality and purity criteria, as well as that they contain the information needed for an adequate assessment of the safety of the cosmetic product.

INCI	Chemical name	Molecular formula	Molecular weight	Water solubility	Vapour pressure (Pa)	log Pow
STEARIC ACID	Stearic acid	C18H36O2	0	< 0.05 mg/L	< 8.54E-06	8.23
GLYCERYL STEARATE SE	1,3-dihydroxypropan-2-yl octadecanoate; 2,3-dihydroxypropyl octadecanoate	C21H42O4	358.56	< 1 mg/L	< 0.0001 Pa at 20 °C	6.1
CETYL ALCOHOL	Hexadecan-1-ol	CH3(CH2)15OH	242.44	<1.0 mg/L at 23 °C, 0.024 mg/L at 25 °C (literature)	N/A	6.7; 7.3 at 23 °C
SYNTHETIC BEESWAX	n/a	n/a	N/A	Insoluble	N/A	N/A
PHENOXYETHANOL	2-Phenoxyethanol	C8H10O2	138.17	26.7 g/l at 20 °C	N/A	log Pow = 1.16
ETHYLHEXYLGLYCERIN	1,2-Propanediol, 3-(2-ethylhexyloxy)	C11H24O3	204.31	Limited solubility in water (cca. 0.1%); highly soluble in organic solvents, such as alcohols, glycols, and glycol ethers.	N/A	log POW = 2.4
PRUNUS AMYGDALUS DULCIS OIL	N/A	N/A	N/A	Insoluble	N/A	N/A
AQUA	Water	H2O	18.02 g/mol	N/A	N/A	N/A
CARBOMER	Polyacrylic acid	N/A	7×105 to 4×109 Da	soluble	N/A	N/A
GLYCERIN	Glycerol	C3H8O3	92.09	Completely miscible with water at 25 °C. The compound is also completely miscible with methanol, ethanol, and the isomers of propanol, butanol, and pentanol.	N/A	log POW = -1.75 at 25 °C
PANTHENOL	2,4-dihydroxy-N-(3-hydroxypropyl)-3,3-dimethylbutanamide	C9H19NO4	205.25	509 g/L	5.6 E-05 Pa at 25 °C	-1.06
UREA	urea	CH4N2O	60,06 g/mol	624 000 mg/L	1.2 x 10 ⁻⁵ mmHg (0.0016 Pa) at 25 °C.	-1.73
POTASSIUM SORBATE	potassium (2E,4E)-hexa-2,4-dienoate	C6H7O2K	150.22 g/mol	1.95 g/L	0 hPa	-1.72
SODIUM BENZOATE	sodium benzoate	C7H5O2Na	144.11	556 g/l	0.0011hPa	-2.269
TRIETHANOLAMINE	2-[bis(2-hydroxyethyl)amino]ethan-1-ol	C6H15NO3	149.2 g/mol	>1000 g/l at 20 °C	< 0.0003 hPa at 21 °C	-2.3

CALENDULA OFFICINALIS FLOWER EXTRACT	0	0	N/A	soluble, insoluble	59.3 mm Hg (ethanol extr)	N/A
CHELIDONIUM MAJUS EXTRACT	N/A	N/A	N/A	soluble	N/A	N/A
ACHILLEA MILLEFOLIUM EXTRACT	N/A	N/A	N/A	soluble	N/A	N/A
GALIUM VERUM EXTRACT	N/A	N/A	N/A	soulble	N/A	N/A
HYPERICUM PERFORATUM EXTRACT	N/A	N/A	N/A	N/A	N/A	N/A
QUERCUS ROBUR BARK EXTRACT	N/A	0	N/A	soluble	N/A	N/A
SALVIA OFFICINALIS LEAF EXTRACT	N/A	N/A	N/A	Insoluble	N/A	N/A
ALCOHOL	ethanol	C2H6O	46.07	789 000 mg/L, completely miscible in water	5 726 Pa	-0.35
AQUA	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
TOCOPHERYL ACETATE	3,4-Dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-benzopyran-6-yl acetate	C31H52O3	472.7	Insoluble in water	N/A	log POW = 12 (estimated)
RETINYL PALMITATE	(2E,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraen-1-yl hexadecanoate	C36H60O2	N/A	The substance appears insoluble in water. Results for water solubility <1mg/l, out of experimental data and for calculated data = 2.8E-11mg/l at 25°C.	extrapolated vapour pressure 5.1E-7 hPa at 20°C	15.51
SALVIA OFFICINALIS OIL	N/A	N/A	N/A	Insoluble	N/A	N/A
EUCALYPTUS GLOBULUS LEAF OIL	N/A	N/A	N/A	2.65 mg/L	253 Pa	4.35
MENTHA PIPERITA OIL	N/A	N/A	N/A	12.3 mg/L	200 Pa	4.38
EUGENIA CARYOPHYLLUS BUD OIL	N/A	N/A	N/A	insoluble	N/A	N/A
THYMUS VULGARIS FLOWER/LEAF OIL	N/A	N/A	0	Insoluble	N/A	N/A
ILLICIAM VERUM LEAF OIL	N/A	N/A	N/A	Insoluble	N/A	N/A
CITRUS LIMON PEEL OIL	0	0	0	0	0	0

b) Physical and chemical characteristics of the cosmetic product

Parameter	Value
Appearance	Cream
Colour	Beige
Odour	Pleasant
pH	6.7
Density	N/A
Viscosity	N/A

c) Cosmetic product's stability

A laboratory study of stability and compatibility was performed, and the study protocol and results are attached to this document.

Stability study protocol description:

- Accelerated test, in original packaging, for 12 weeks, on +5C, room temperature (25C) and +40C,.
- Centrifugation measurement
- Measurement of organoleptic parameters (color, odor), pH, viscosity and phase separation were measured periodically
- Microbiology testing at the beginning and at the end of the test period

Based on the results obtained, it is concluded that the product DERMOTIN BIOWART is stable and compatible with the packaging under reasonably applicable conditions of use and storage on 5C, 25C and 40C.

3) Microbiological quality

a) Microbiological quality of raw materials

Raw materials used for GINGERIA GROW HEMORODIL are followed with technical documentation (SDS, TDS, CoA, etc.) provided by the supplier which is available and attached to this document. The technical documentation was reviewed, based on which it was concluded that all the ingredients of the cosmetic product represent a satisfactory microbiological quality. The production of the cosmetic product is carried out in accordance with the principles of good manufacturing practice, which excludes the possibility of microbiological contamination during the production process.

b) Microbiological quality of the cosmetic product

Quantitative and qualitative limits are based on the European Standard EN ISO 17516:2014 Cosmetics – Microbiology – Microbiological limits. The European Standard EN ISO 17516:2014 was approved by CEN on 9 August 2014 and at present is widely used 149 by the cosmetics industry as international standard as given (SCCS Notes of Guidance 11th revision):

Types of microorganism	Products specifically intended for children under three years of age, the eye area or the mucous membranes	Other products
Total Aerobic Mesophilic Microorganisms (Bacteria plus yeast and mould)	$\leq 1 \times 10^2$ CFU per g or ml ^a	$\leq 1 \times 10^3$ CFU per g or ml ^b
<i>Escherichia coli</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
<i>Pseudomonas aeruginosa</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
<i>Staphylococcus aureus</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
<i>Candida albicans</i>	Absence in 1 g or 1 ml	Absence in 1 g or 1 ml
Due to inherent variability of the plate count method, according to USP Chapter 61 or EP Chapter 2.6.12, Interpretation of results, results considered out of limit if a > 200 CFU/g or ml, b > 2 000 CFU/g or ml. NOTE When colonies of bacteria are detected on Sabouraud Dextrose agar, Sabouraud Dextrose agar containing antibiotics may be used.		

Microbiological testing of the cosmetic product DERMOTIN BIOWART was conducted and the results are proving the compliance of the product with the limits named above for the category „Other products“. The laboratory results report is attached to this document.

Challenge test: The efficacy of preservation system used for cosmetic product DERMOTIN BIOWART was conducted using method ISO 11930:2019 - Cosmetics - Microbiology - Evaluation of the antimicrobial protection of a cosmetic product. The result obtained confirmed the efficacy of the antimicrobial protection of a cosmetic product. The laboratory results are attached to this document.

4) Impurities, traces, information about the packaging material

a) Impurities and traces in the raw materials

Raw materials used for the manufacturing of the cosmetic product DERMOTIN BIOWART are well-known chemical substances and mixtures that have long been used as raw materials in the cosmetic industry.

Based on the technical documentation (SDS, TDS, CoA, etc.) provided by the suppliers of used raw materials, the presence of impurities and traces was determined for which the Margin of Safety (MoS) was calculated using the Threshold of Toxicological Concern (TTC) calculation:

INCI	Cramer Class	NOAEL	SED	MoS
Ethyl Acetate	I	0.03	0.000132245	226.851
Cyclohexane	I	0.03	1.14996E-05	2608.786
Formaldehyde	III	0.000025	1.15817E-05	2.15857

The Threshold of Toxicological Concern (TTC) calculation is a scientifically acceptable method for assessing the risk of systemic toxic effects that may arise from the use of chemical substances present in very small quantities. This evaluation method is considered acceptable in case of lack of adequate toxicological data necessary to assess the risk of intentionally using additional cosmetic ingredients as well as the impurities present in them.

The Cramer class was determined by using software TOXTREE.

Results obtained for the MoS indicate safe exposure to ingredients with identified impurities.

b) Heavy metals content

Heavy metals content of the cosmetic product DERMOTIN BLOWART was determined for Pb, Cd, As, Cr, Hg, and Ni. The results are given as less than 0.5mg/kg for each component which is considered acceptable. The laboratory results report is attached to this document.

Cosmetic product DERMOTIN BLOWART is produced in Republic of Serbia where heavy metals content determination is mandatory. Results obtained are acceptable according to the limits set by the cosmetic regulation in Republic of Serbia.

c) The relevant characteristics of packaging material

Cosmetic product DERMOTIN BLOWART was packed in a white plastic cosmetic container with a lid made from the same material. The container is being placed in a cardboard box. Elements used for packaging described are well known and are being widely used as packaging for pharmaceutical and cosmetic products for a long time.

The packaging supplier submitted the technical documentation for each element that is attached to this document and is describing the used materials.

5) Normal and reasonably foreseeable use

Cosmetic product DERMOTIN BLOWART is intended to be used as a anal area skin care cream without subsequent washing after application.

Instructions for use written on the cosmetic product:

„Apply the cream to anal area 1-2 times daily.“

Warnings and precautions written on the cosmetic product:

„Sensitivity to any ingredient. Avoid contact with eyes. If the contact occurs, wash with plenty water. In case the irritation does stop ask for a doctor’s advice“.

The cosmetic product DERMOTIN BLOWART is presented in a way that clearly communicates its purpose. Precise and clearly understandable instructions are given, based on which it is considered that it cannot be misused.

6) Exposure to the cosmetic product

Duration of product application	Leave-on
Body part intended for application	Parts of skin
Reasonably foreseeable application on other body parts	N/A
Duration and frequency of application	2 times a day
Relative daily exposure (mg/kg bw/day)*	41.07
Target population	Adults
Retention factor	1

*estimated according to the SCCS Notes of Guidance for the testing of cosmetic ingredients and their safety evaluation (12th revision)

7) Exposure to the cosmetic products ingredients

Systemic Exposure Dose (SED) was calculated using the formula:

$$\text{SED (mg/kg bw/day)} = A \text{ (mg/kg bw/day)} * C \text{ (\%)/100} * DA \text{ (\%)/100}$$

Where parameters are:

A (mg/kg bw/day) – Relative daily exposure

C (%) – Maximum concentration of the ingredient in the final product

DA (%) – Dermal absorption

Dermal absorption (DA) and no observed adverse effect level (NOAEL) values were determined by detailed analysis of the toxicological profiles of substances that are given in the part 8. If oral bioavailability and dermal absorption values were not known, the default value of 50% was used.

Margin of safety (MoS) was calculated using the formula:

$$\text{MoS} = \text{NOAEL} / \text{SED}.$$

INCI naziv sastojka	Maksimalna koncentracija sastojka u proizvodu (%)	SED (mg/kg bw/day)	NOAEL (mg/kg bw/day)	MoS
STEARIC ACID	1.97	0.4045395	500	411.9911
GLYCERYL STEARATE SE	1.28	0.262848	500	634.08
CETYL ALCOHOL	1.46	0.299811	1000	1111.812
SYNTHETIC BEESWAX	1.1	0.225885	400	1770.813
PHENOXYETHANOL	0.91	0.1868685	N/A	N/A
ETHYLHEXYLGLYCERIN	0.2275	0.046717125	25	535.1357
PRUNUS AMYGDALUS DULCIS OIL	0.91	0.1868685	N/A	N/A
AQUA	71.432	12.193683	N/A	N/A
CARBOMER	0.27	0.0554445	N/A	N/A
GLYCERIN	1.83	0.3757905	100	266.1057
PANTHENOL	0.91	0.1868685	100	178.3786
UREA	0.91	0.1868685	125	668.9196
POTASSIUM SORBATE	0.09	0.1868685	N/A	N/A
SODIUM BENZOATE	0.18	0.036963	N/A	N/A
TRIETHANOLAMINE	0.75	0.1540125	50	324.649
CALENDULA OFFICINALIS FLOWER EXTRACT	2.1743	0.446492505	200	447.9359
CHELIDONIUM MAJUS EXTRACT	2.1743	0.446492505	607	1359.485
ACHILLEA MILLEFOLIUM EXTRACT	1.1511	0.236378385	400	564.0674
GALIUM VERUM EXTRACT	1.1511	0.236378385	30	126.9152
HYPERICUM PERFORATUM EXTRACT	1.1511	0.236378385	13	54.99657*
QUERCUS ROBUR BARK EXTRACT	1.1511	0.236378385	100	423.0505
SALVIA OFFICINALIS LEAF EXTRACT	1.1511	0.236378385	N/A	N/A
ALCOHOL	10.232	2.1011412	1200	571.1182
AQUA	10.232	2.1011412	N/A	N/A
TOCOPHERYL ACETATE	0.09	0.0184815	180	3246.49
RETINYL PALMITATE	0.09	0.0036963	1.43	386.8734
SALVIA OFFICINALIS OIL	0.06	0.012321	5	135.2704
EUCALYPTUS GLOBULUS LEAF OIL	0.06	0.012321	300	8116.224

MENTHA PIPERITA OIL	0.06	0.012321	40	3246.49
EUGENIA CARYOPHYLLUS BUD OIL	0.06	0.012321	2.5	202.9056
THYMUS VULGARIS FLOWER/LEAF OIL	0.06	0.012321	250	6763.52
ILLCIUM VERUM LEAF OIL	0.06	0.012321	5	405.8112
CITRUS LIMON PEEL OIL	0.06	0.012321	170	13797.58

*NOAEL from human data.

According to the guidelines of the World Health Organization and the Scientific Committee on Consumer Safety (SCCS), a Margin of Safety (MoS) value of at least 100 is considered a sufficient indicator of safe use of the substance.

8) Toxicological profile of the substances

INCI	ALCOHOL
Presence in the annexes	0
Description	According to the International Cosmetic Ingredient Dictionary and Handbook (Gottschalk and McEwen 2004), Alcohol is the name given by convention to ethanol as used in cosmetics, and Alcohol Denat. is the name given to denatured Alcohol for use in cosmetics. Alcohol Denat. is prepared by adding one or more denaturing agents to potable Alcohol, making it unfit for beverage or internal human medicinal use. Alcohol may be completely denatured (CD Alcohol) or specially denatured (SD Alcohol) depending on the type(s) of denaturant(s) used.
Oral toxicokinetics	Following any route of intake resulting in an elevated blood ethanol level . Inhalation: In a study to determine the fractional uptake of ethanol during inhalation exposure, human volunteers were exposed to ethanol vapour concentrations of 10 -12mg/l (6000ppm) for periods of 1 hour. Experiments were done with and without a prior dose of ethanol by the oral route. The fractional absorption was typically around 55% (variation less than 10% within and between individuals) and was found to be not detectably affected by variations in tidal volume, breathing frequency nor the existing blood ethanol concentration. The concentration of ethanol in expired air did not fall below 30%.
Inhalation toxicokinetics	In a study to determine the fractional uptake of ethanol during inhalation exposure, human volunteers were exposed to ethanol vapour concentrations of 10 -12mg/l (6000ppm) for periods of 1 hour. Experiments were done with and without a prior dose of ethanol by the oral route. The fractional absorption was typically around 55% (variation less than 10% within and between individuals) and was found to be not detectably affected by variations in tidal volume, breathing frequency nor the existing blood ethanol concentration. The concentration of ethanol in expired air did not fall below 30%.
Dermal absorption	In a study to assess the skin penetration potential of ethanol, an in vitro study was carried out using excised pig skin and radiolabelled ethanol. Ethanol penetration was greater under occlusion conditions than non-occlusive

	conditions, as might be expected. Absorption rates were around 21% and 1% of applied doses respectively.
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	In a guideline acute oral toxicity study, the LD50 was determined to be 10470 mg/kgbw when dosed as a 95% solution in water. Dermal: LD50 17 100 mg/kg bw. Inhalation: In an acute inhalation study that approximated to guideline, ethanol vapour was found to be of very low acute toxicity to male and female rats. The LC50 (4hr) was established to be around the 117 -125mg/l and the LC0 around 62mg/l. It is worthy of note that the LD50 is well above the lower explosive limit (LEL).
Repeated dose studies	Oral treatment of rats for 14 weeks with 0, 5, 10 or 20 mg/Kg of a mixture containing 16.25% USP ethanol, The NOAEL for the study is determined at 10 ml/Kg. Most data available on ethanol is via the oral route of exposure. Much is at high doses which limits its value to risk assessment of ethanol as a chemical substance. From the data available, it is possible to surmise that ethanol is of repeat dose low toxicity by the oral route, with a lowest reported NOAEL in animals of 2400 mg/kg for rats.
Mutagenicity/genotoxicity	OECD Guideline 471 (Bacterial Reverse Mutation Assay): negative. In vivo mammalian germ cell study: cytogenicity / chromosome aberration: Ethanol is unlikely to be a dominant lethal mutagen, at least up to the maximum tolerated dose.
Carcinogenicity	Rats: No effect level >3000mg/kg. Mice (B6C3F1): Females: NOAEL>44000mg/kg (cancer), Males: NOAEL>4250mg/kg (based on historic control data), Males BMDL10=1400mg/kg (based on concurrent control data). Overall, there is no significant evidence to warrant a classification of ethanol for cancer in the context of the relevant classification and labelling regulations for chemical substances.
Reproductive toxicity	Overall, ethanol in drinking water at concentrations up to 15% (equivalent to 20.7 g/kg/day) had no demonstrable effect on fertility in this two-generation study. Groups of female Sprague-Dawley rats were given liquid diets with 15, 25, or 36% ethanol-derived calories (E15, E25, and E36 groups, respectively), or without ethanol (pair-fed isocaloric (PF15, PF25, PF36) or ad libitum (C) control) for 3 weeks prior to mating, and throughout 21 days of gestation. Prenatal ethanol exposure affected fetal skeletal ossification at exposure levels lower than those required to affect fetal body weight and length, although the significance of these changes for long-term bone health is unknown. No statistically significant effects on fetal growth or ossification were seen in the fetuses of dams ingesting about 5.2 g ethanol/kg bw/day from 3 weeks prior to mating to GD 21.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/16105/7/10/2 2. OECD SIDS, Ethanol, Assessment Report https://hpvchemicals.oecd.org/ui/handler.axd?id=2602cc56-d998-4e67-bd78-5454ef3f8f9a
POD (NOAEL)	2400mg/kg bw/day

INCI	GLYCERIN
Presence in the annexes	0

Description	Glycerol is widely distributed in food as a natural constituent, and it has undergone review and approval for use as both a direct and indirect food additive and is GRAS by US FDA. Animal studies involving single or repeated exposure by various routes indicated a low order of toxicity, the possible target sites (particularly following repeated oral administration) being the kidney and gastrointestinal tract. Glycerol appears to be of generally low oral toxicity in humans.
Oral toxicokinetics	Glycerin is rapidly absorbed in the intestine and the stomach, distributed throughout the extracellular fluids through much of the body. It is mostly metabolized by the liver and kidneys with the remainder excreted in urine. Free glycerin is naturally present in humans, primarily in plasma.
Inhalation toxicokinetics	The NOAEC in rats for local and systemic effects after 13-week exposure is 662 mg/m ³ .
Dermal absorption	0.1
Skin irritation	Not dermally irritating to rabbits when applied at concentrations up to 100% to up to 30% of the body surface. Mild dermal irritant at 100% in guinea pigs.
Eye irritation	Undiluted glycerin was not irritating when administered to the eyes of human subjects. There was a strong burning and stinging sensation, with tear production but no injury was observed.
Skin sensitisation	A moisturizer containing 65.9% glycerin was not sensitizing to human subjects. Natural and synthetic glycerin were not sensitising to white male guinea pigs at 0.1%.
Acute toxicity	The reported oral LD50 of glycerin ranged from 2530-58400 mg/kg in rats, 4090-38000 mg/kg in mice, 27000 mg/kg in rabbits, and 77500 mg/kg in guinea pigs. The dermal LD50 of glycerin in rats was reported to be > 21900 mg/kg and >18700 mg/kg in rabbits. The approximate Lt50 for rats was determined to be 423 min for exposure to glycerin vapors at 11.0 mg/L. The intraperitoneal LD50 of glycerin in rats ranged from 4420-10100 mg/kg and 8600-9500 mg/kg in mice. The subcutaneous LD50 of glycerin was 100 mg/kg in rats and ranged from 91-10000 mg/kg in mice. The intravenous LD50 of glycerin ranged from 5200-6600 mg/kg in rats, 4250-6700 mg/kg in mice, and 53000 mg/kg in rabbits.
Repeated dose studies	In the best available dietary study, groups of 22 rats (Long-Evans)/sex/treatment received 5, 10 and 20% glycerol (natural or synthetic) in their diet (males 2000, 4000 and 8000 mg/kg bw; females 2500, 5000 and 10000 mg/kg bw) for 2 years. Although the results were not described in detail, based on this limited dietary study it can be concluded that no adverse effects were observed at up to 10,000 mg/kg bw. Dermal: The effects following repeated dermal application of glycerin was examined. There were no effects noted in rabbits dosed 8 hours/day, 5 days/week for 45 weeks with dose levels as high as 4.0 ml/kg. Using a density of 1.2611 g/cm ³ at 20 °C, a dose of 4.0 ml/kg corresponds to 5040 mg/kg/day.
Mutagenicity/genotoxicity	Glycerin was not genotoxic in multiple Ames tests using multiple strains of <i>S. typhimurium</i> at concentrations up to 50 mg/plate. It was not genotoxic in a cytogenetic assay, X-linked HGPRT, sister chromatid exchange assay, unscheduled DNA synthesis assay, and chromosome aberration test at concentrations up to 1.0 mg/mL.
Carcinogenicity	Glycerin administered in the feed of rats at doses up to 20% in feed for 1 year or up to 10 g/kg for 2 years did not increase the incidence of tumors. Orally administered glycerin, in concentrations up to 5%, had a potentiating effect on the carcinogenicity of 4NQO in mice.
Reproductive toxicity	No convincing evidence of reproductive effects have been seen in rats treated orally or dermally, or in mice fed glycerol during pregnancy. Glycerin is transferred across the placenta in small amounts. May cause some adverse reproductive effects based on animal data, but there was no evidence of

	teratogenicity. No effects on fertility and reproductive performance were observed in a two generation study with glycerol administered by gavage (NOAEL 2000 mg/kg bw/day).
Photosensitivity	N/A
Endocrine effects	N/A
Reference	(1) CIR Expert Panel. Safety assessment of glycerin as used in cosmetics - final report (January 14,2015). (2) ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/14481/7/6/1
POD (NOAEL)	reproduction 2000, oral 10000, dermal 5040

INCI	POTASSIUM SORBATE
Presence in the annexes	V/4
Description	Potassium sorbate (E 202) is authorised for use as a food additives in the European Union. It is present in Annex V of 1223/2009 Directive with max. Allowed concentrations in ready for use preparations of 0.6% (acid).
Oral toxicokinetics	Sorbic Acid as well as Potassium Sorbate is rapidly and completely metabolized in the animal. The extrapolation of the metabolism of sorbic acid to potassium sorbate is considered not to be restricted in anyway, since the determination of potential toxicity is on the "sorbate" anion.
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	not irritating
Eye irritation	irritating
Skin sensitisation	not sensitising
Acute toxicity	Sorbic Acid and Potassium Sorbate show a very low toxicity in rats and mice in acute oral toxicity studies. The median lethal dose (LD50) of the Sorbic Acid is 10.50±1.96 g/kg body weight in the rat, while the LD50 of the Potassium salt has been determined at 3.2-10.5 g/kg body weight in the rat.
Repeated dose studies	EFSA Panel established a group ADI expressed as 11 mg sorbic acid/kg bw per day for sorbic acid (E 200) and its potassium salt (E 202).
Mutagenicity/genotoxicity	Sorbic Acid and its salts are non-mutagenic and non-clastogenic in vitro and in vivo.
Carcinogenicity	Potassium sorbate and sorbic acid are not carcinogenic.
Reproductive toxicity	An EOGRS (extended one generation reproductive toxicity study) compliant with GLP was performed in CD/Crl:CD rats following OECD TG 443 regarding cohorts 1A and 1B (reproductive toxicity). In all generations, neither test item-related mortality nor effect on clinical signs in the animals and their pups were observed.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. EFSA Panel "Opinion on the follow-up of the re-evaluation of sorbic acid(E200) and potassium sorbate (E202) as food additives" https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2019.5625 2. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/11008/7/9/1
POD (NOAEL)	11mg/kg bw/day

INCI	Stearic Acid
Presence in the annexes	None
Description	Stearic acid C18. Fatty acids are typically unbranched, long chain organic acids with different chain length. They are found in all living organism fulfilling three fundamental roles. Besides their function as part of molecules like phospholipids and glycolipids important for the cell-structure, they are often precursors of signalling molecules such as prostanooids in animals or phytohormones in plants. The third and best understood role of fatty acid is their role as an energy source, particularly in higher animals and plants.
Oral toxicokinetics	Taken together the experimental and calculated data show that fatty acids are almost completely absorbed after oral intake, whereas only limited dermal uptake has to be expected.
Inhalation toxicokinetics	LC50 >0.1521 mg/L
Dermal absorption	poorly absorbed through skin, with a measured rate of less than 1 % after 24-hours exposure
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	LD50 >6000 mg/kg bw
Repeated dose studies	Combined repeated dose and reproductive/developmental toxicity screening test, NOAEL 1000mg&kg bw&daz, duration unknown
Mutagenicity/genotoxicity	All available data on genotoxicity showed that fatty acids are not genotoxic.
Carcinogenicity	N/A
Reproductive toxicity	Combined repeated dose and reproductive/developmental toxicity screening test, NOAEL 1000mg&kg bw&daz, duration unknown
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/15163/7/7/1
POD (NOAEL)	1000mg/kg bw/day, rats, oral, duration unknown

INCI	Glyceryl Stearate SE
Presence in the annexes	None
Description	Monoglyceryl monoester.
Oral toxicokinetics	No data on oral absorption
Inhalation toxicokinetics	N/A
Dermal absorption	Monoglyceryl monoesters can act as penetration enhancers.
Skin irritation	not irritating
Eye irritation	Slight to moderate effects on the conjunctivae (redness and chemosis) and slight effects on the iris (congestion) were observed in all animals 1 h after test substance instillation. In a single animal, effects on the iris persisted up to 48 h and reversed thereafter. No effects on cornea were observed in any animal tested. Not irritating
Skin sensitisation	not sensitising
Acute toxicity	> 5 000 mg/kg bw
Repeated dose studies	NOAEL, 90 days, rat> corresponding to doses of 5835 and 5725 mg/kg bw/day in males and females, respectively, as calculated from the reported food consumption and body weights
Mutagenicity/genotoxicity	In vitro> Genotoxicity: negative. Cytotoxicity> Not cytotoxic
Carcinogenicity	There is no evidence that the members of the Glyceride category induce gene mutations in bacteria or chromosome aberrations in mammalian cells, as the

	results of all genotoxicity studies were consistently negative. Furthermore, in the available repeated dose toxicity studies and developmental studies, no substance-related increases in the incidence of hyperplasia or pre-neoplastic lesions were observed. Justification for selection of carcinogenicity via oral route endpoint: No carcinogenicity study is required, since the substance is not mutagenic and no hyperplasia or pre-neoplastic lesions were observed in any of the available studies. The available and relevant studies do not indicate that the members of the Glyceride category fulfil the criteria for classification as germ cell mutagen or that they are able to induce hyperplasia and/or pre-neoplastic lesions.
Reproductive toxicity	NOAEL, parental fertility 1 000 mg/kg bw/day (actual dose received). Embryotoxic / teratogenic effects: no effects
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/regISTRATION-dossier/-/registered-dossier/2133/7/2/2 2. CIR Exper Panel, Amended Safety Assessment of Monoglyceryl Monoesters as Used in Cosmetics
POD (NOAEL)	1000 from reproduction toxicity, duration unknown

INCI	Cetyl Alcohol
Presence in the annexes	0
Description	It is a 16-carbon alcohol, known also as 1 -hexadecanol and n-hexadecyl alcohol. Cetyl Alcohol is used as an emollient to prevent drying and chapping of the skin because of its water-binding property. Because Cetyl and Stearyl Alcohols have structural similarities, the literature review may be applicable to Cetyl Alcohol. Long-chain aliphatic alcohols (C ₁₂ and C ₁₆) induced minimal ocular and skin irritation but no sensitization or comedogenicity in rabbits; no mutagenic effects were noted in the Ames assay. In a subchronic percutaneous toxicity study, a product formulation (C ₁₂ -alcohol content) induced erythema and mild desquamation. Clinical studies of long-chain alcohols (C ₁₂ and C ₁₆) indicated a low order of skin irritation and sensitization. Also, results were negative in clinical phototoxicity and photosensitization studies of products containing these alcohols.
Oral toxicokinetics	No bioaccumulation potential based on study results. Following skin application of lauryl alcohol about 2.84 % of the administered dose was absorbed. Of this absorbed dose >90% was excreted in expired air (CO ₂). A similar trend was observed with the other alcohols tested, including hexadecan-1-ol. Absorption decreased with increasing carbon chain length and was affected by solvent and concentration. Good absorption: Approximately 15% of the alcohol was unchanged during its passage through the mucosal cells of the small intestine;
Inhalation toxicokinetics	N/A
Dermal absorption	Of a dose of 5% 1-14C-cetyl alcohol applied to the skin of nude mice for 24 hours under occlusive conditions, 1% was absorbed.
Skin irritation	Following a 4 hour semi-occlusive exposure of Kalcol 6098 to rabbit skin there was no evidence of skin irritation between 24 and 72 hours after patch removal. Kalcol 6098 is not a skin irritant according to EU or GHs criteria.

Eye irritation	Kalcol 6098 is not an eye irritant according to EU or GHS criteria. The result is a read across from hexadecanol (CAS 36653-82-4).
Skin sensitisation	In a reliable study performed using a protocol similar to OECD guideline 406, Dobanol 25 was not sensitising to the skin of guinea pigs after a topical challenge with 2.5% when tested using the method of Magnusson and Klighman. (1977)
Acute toxicity	The rat oral LD50 for Kalcol 6098 is >2000 mg/kg. At this dose level there were no signs of toxicity. The rat 1 hour inhalational LC50 for tetradecan-1-ol is >1.5 mg/l. There were no signs of toxicity and findings at gross necropsy were unremarkable. The rabbit dermal LD50 (24 hour occluded) for Alfol 14 was approx. 8000 mg/kg.
Repeated dose studies	For hexadecan-1-ol, oral NOAELs were > 4257 and > 4567 mg/kg bw/day in male and female rats respectively in a 90-day repeated dose toxicity test in which a somewhat limited range of endpoints was evaluated (Scientific Associates 1966a). the systemic No-Observed-Adverse-Effect-Level following dermal administration of fatty alcohol blend for a minimum of 90 days was considered to be 1000 mg/kg/day (the highest dose tested).
Mutagenicity/genotoxicity	In a reliable study, performed according to OECD guideline 471, the C16 alcohol Kalcol 6098 did not increase the reverse mutation rate in histidine dependent bacterial strains of Salmonella typhimurium in the presence or absence of metabolic activation at concentrations up to 5000 µg/plate. This concentration was not cytotoxic. Fatty alcohol blend has been tested according to OECD 474 and under GLP. Male and female mice were dosed with 500, 1000 and 2000 mg/kg bw. No increase in the number of micronucleated PCE was observed (1000 PCE scored per animal). It is concluded that the test substance is negative for the induction of micronuclei under the conditions of the test. No toxicity to bone marrow or general toxicity was observed.
Carcinogenicity	the weight of evidence across the category suggests that hexadecan-1-ol is not carcinogenic. This is based on a large set of various types of repeated dose studies across the category which do not offer any evidence of treatment-related induction of hyperplasia / pre-neoplastic lesions for hexadecan-1-ol or structurally related alcohols (though reporting is limited in many cases). In addition, hexadecan-1-ol does not have any genotoxic effects.
Reproductive toxicity	In a reliable study conducted to the draft OECD guideline 422, a parental NOAEL of > 2000 mg/kg bw/day (highest dose tested) was determined for male and female rats. No adverse effects were observed on reproductive parameters and the NOAEL for reproductive and developmental effects was also > 2000 mg/kg bw/day. In a reliable study, development was assessed as part of a combined repeat dose and reproductive/developmental toxicity study, conducted according to draft OECD guideline 422. The NOAEL for maternal and foetotoxicity in rats was 2000 mg/kg bw/day (highest dose level). There was no evidence of teratogenicity from the limited examination of the pups that was carried out. The result is read across from octadecan-1-ol (CAS 112-92-5).
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. Publication: Percutaneous absorption of aliphatic compounds, Iwata Y, Moriya Y, Kobayashi T, Cosmet. Toiletries 102(2): 53-68 (1987). 2. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/15942/1/1 3. CIR Expert Panel, Final Report on the Safety Assessment of Cetearyl Alcohol, Cetyl Alcohol, Isostearyl Alcohol,

	Myristyl Alcohol, and Behenyl Alcohol
POD (NOAEL)	2000 from Reproduction toxicity as worst case, oral high, duration unknown

INCI	Cera Alba
Presence in the annexes	0
Description	Beeswax is a GRAS food ingredient. In cosmetics, it is used in baby products, bath and fragrance preparations, eye and facial makeup, hair, skin, nail, shaving and suntan preparations, at concentrations of less than 0.1% to greater than 50%. A total content of 27 to 40% monoesters, 9 to 23 %hydroxymonoesters, 7 to 16% diesters, 3.9 % hydroxydiesters, 11 to 28% hydrocarbons, 1 to 18% saturated, unbranched free fatty acids, 4 to 8% of other substances and < 0.3% free fatty alcohols have been reported in beeswaxes from different bees species. Experimental biochemical and toxicological studies on beeswax as such are lacking. However, the Panel considered that the safety of beeswax could be evaluated on the basis of published toxicological studies on its main constituents. An EFSA Committee considered that the data on beeswax itself were insufficient to establish an ADI, but concluded that the safety of beeswax could be assessed, based on available scientific literature on the main constituents of beeswax and plant waxes showing chemical structural similarities to beeswax. They could not reach a conclusion about the potential allergenicity of beeswax as the available information was very limited. However, the committee concluded that the use of beeswax as an additive for the existing food uses is not of safety concern. (EFSA 2007).
Oral toxicokinetics	Oral: low.
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	mild acute skin irritation
Eye irritation	Mild irritation
Skin sensitisation	N/A
Acute toxicity	Acute oral doses of 5 g/kg Beeswax produced no toxicity in rats.
Repeated dose studies	A cleansing cream formulation containing 13% Beeswax was used in a sixweek dermal toxicity study using 20 rats. A 0.3 g/kg dose of the test material was applied to the shaved skin of each of 10 rats, once per day, five days per week, ASSESSMENT: CANDELILLA WAX, CARNAUBA WAX, JAPAN WAX, AND BEESWAX 29 for a total of 30 applications. Ten other rats were used as untreated controls. Observations were made throughout the test period for general appearance, behavior, and toxicologic signs. Body weights were obtained weekly and blood samples were taken before the end of the study. All animals survived the study, and had normal appearance and behavior. External appearance of the skin at the site of application did not differ from the skin of untreated controls. Gross and microscopic examinations detected no abnormalities, and it was concluded that the formulation produced no systemic toxic effects.
Mutagenicity/genotoxicity	Beeswax was not mutagenic in tests using Salmonella typhimurium or Saccharomyces cerevisiae with and without metabolic activation.(g3
Carcinogenicity	Threads impregnated with Beeswax (controls) or 3-methylcholanthrene (MC) plus Beeswax were implanted in the cervixes of six-week-old virgin female

	C341HeJ mice. Epithelialization of the cervical canal occurred after two and four weeks with both Beeswax and Beeswax plus MC. Invasive carcinoma developed in mice treated with Beeswax plus MC after four weeks of treatment, but not in the control (Beeswax-treated) animals.(91) In a similar experiment, Beeswax and Beeswax plus MC-impregnated thread were implanted into the cervixes of mice. In Beeswax-treated mice, and invasive squamous carcinoma was found in one of the 35 mice tested after four and six weeks; MC plus Beeswax induced 58% carcinoma of the cervix in treated mice.”)
Reproductive toxicity	N/A
Photosensitivity	A mascara formulation containing 10.0% Beeswax was tested for photosensitivity on 68 panelists. Applications of 0.1 ml/cm ² were made for 24 h, evaluated for irritation, and then irradiated with a 150 W quartz bulb lamp with a continuous emission of 300-370 nm at a distance of 12 in from the test site. Scoring of the irradiated sites were made after 48 h, and the process was repeated six times. After a 10-day nontreatment period, a challenge application was applied to an adjacent, untreated site for 24 h. This new site was then irradiated. After 72 h, none of the 68 subjects developed photosensitivity.(104
Endocrine effects	N/A
Reference	1. CIR Expert Panel, Final Report on the Safety Assessment of Candelilla Wax, Carnauba Wax, Japan Wax, and Beeswax 2. https://efsa.onlinelibrary.wiley.com/doi/pdf/10.2903/j.efsa.2007.615
POD (NOAEL)	300mg/kg bw/day from 6weeks repeated dose study for a 13% beeswax cream. NOAEL for beeswax could be taken as 800 as worst case (it is 13% in the cream)

INCI	Phenoxyethanol
Presence in the annexes	V/29
Description	European Cosmetics Regulation Annex V (Line 29) lists phenoxyethanol as an approved preservative in the EU at concentrations up to 1%. In terms of Regulation (EC) No 1272/2008 (CLP) the following hazard classes are assigned to this ingredient (EU harmonized classification): Acute Tox. 4, Eye Irrit. 2.
Oral toxicokinetics	Tests with rats show that more than 75% and up to 99% of the phenoxyethanol after either oral or dermal exposure can be found unchanged in the urine together with small quantities of two substances to which the phenoxyethanol has metabolized. One of the metabolism products is phenoxyacetic acid.
Inhalation toxicokinetics	N/A
Dermal absorption	Rapidly absorbed through the rat skin.
Skin irritation	Not skin irritating (humans); slightly irritating (animals)
Eye irritation	Irritating (rabbit)
Skin sensitisation	Not sensitizing (human, animal)
Acute toxicity	Oral LD50 (rat): 1386 - 4013 mg/kg; Dermal LD50 (rat): 14300 mg/kg
Repeated dose studies	Dermal exposure is the usual exposure route for most cosmetic ingredients such as 2-phenoxyethanol and more relevant than oral exposure, which is often used as a surrogate exposure when animal safety studies are conducted with cosmetic ingredients. This view on the limited relevance of oral RDT data and dermal exposure as the relevant exposure route is supported by the toxicokinetic data (section 3.3.9), where an extensive first-pass metabolism after oral exposure of rats was found so that only traces of the haematotoxic agent 2-phenoxyethanol reached the systemic circulation. In contrast, after

	dermal exposure of rats even to low doses of 2-phenoxyethanol, up to 20% of the dose appeared in the blood as the parent compound (Kim et al. 2015, AR 9). Therefore, the 90-day dermal toxicity study in rabbits SCCS/1575/16 Final version of the Opinion on Phenoxyethanol with an adjusted NOAEL of 357 mg/kg bw/day is used as the key study for the MoS calculation.
Mutagenicity/genotoxicity	Nonmutagenic in the Ames test and in the mouse micronucleus test.
Carcinogenicity	Not expected to be carcinogenic.
Reproductive toxicity	Phenoxyethanol has shown damaging impacts on reproduction and developmental toxicity in animal tests with mice. In several reproduction studies with mice the impacts were decreasing body weight on the mice and their progeny as well as increased liver weight at high doses of between 1875 and 4000 mg/kg/day.
Photosensitivity	In clinical studies phenoxyethanol was nonphototoxic.
Endocrine effects	N/A
Reference	(1) Danish Ministry of Environment, EPA. A survey and health assessment of cosmetic products for children. Survey of Chemical Substances in Consumer Products, No. 88 2007. (2) CIR Expert Panel. Final Report on the Safety Assessment of Phenoxyethanol. JACT 9(2):259-277, 1990.
POD (NOAEL)	357 dermal

INCI	Ethylhexylglycerin
Presence in the annexes	0
Description	The safety of ethylhexylglycerin has been assessed by the CIR Expert Panel. The CIR Expert Panel evaluated the scientific data and concluded that this ingredient is safe as used in cosmetics at concentrations up to 8%. Although ethylhexylglycerin is generally considered safe, it has been found to be an irritant to the skin in two studies, causing allergic contact dermatitis.
Oral toxicokinetics	In an acute toxicokinetic study in vivo, the mean absorption of ethylhexylglycerin through the skin of rabbits was 0.02% at approximately 2-hour postapplication, and there were no signs of skin irritation. The amount of ethoxyglycerin in the plasma was below the detection limit at the end of the 4-hour application period.
Inhalation toxicokinetics	N/A
Dermal absorption	Over a range of 3 concentrations (44.65%, 47.15%, and 54.94%) applied to human skin in vitro, mean penetration rates of 2.38, 8.19, and 20.38 µg/cm ² /h were reported. Rabbit The mean absorption through skin in this group was 0.02% at approximately 2 hours after the beginning of exposure.
Skin irritation	Non-irritating in HRIPT (up to 1%); mildly to non-irritating (undiluted, rabbit).
Eye irritation	Undiluted ethylhexylglycerin was severely irritating, but 5% ethylhexylglycerin was mildly irritating, to the eyes of rabbits.
Skin sensitisation	Not sensitizing (up to 50%, guinea pig and local lymph node assay); not sensitizing (0.4-1%, HRIPT). Positive patch test results were reported for dermatitis patients patch tested with ethylhexylglycerin at concentrations up to 10%. However, results were negative for control group.
Acute toxicity	No mortalities or exposure-related toxicological findings were observed in rats dosed orally with undiluted ethylhexylglycerin. Oral LD50 (rat) > 2000 mg/kg
Repeated dose studies	In rats dosed orally with ethylhexylglycerin at doses up to 1500 mg/kg for 28 days, the 100 mg/kg dose was defined as the NOAEL. Ethylhexylglycerin

	administered orally to rats, at doses up to 800 mg/kg/day, in a 13-week study, a NOAEL of 50 mg/kg/day (lowest dose) was derived.
Mutagenicity/genotoxicity	Not genotoxic in Ames tests using multiple strains of <i>S. typhimurium</i> and in the mouse lymphoma assay in vitro. Nonclastogenic in the micronucleus assay in vivo.
Carcinogenicity	N/A
Reproductive toxicity	In a preliminary test, the results of visceral and skeletal examinations in litters of female rats given oral doses of ethylhexylglycerin up to 800 mg/kg/day were negative. The NOAEL for developmental toxicity was considered to be 800 mg/kg/day. In the 1-generation developmental toxicity study in rats, NOEL was 50 mg/kg/day.
Photosensitivity	Not phototoxic or photoallergic (guinea pig).
Endocrine effects	N/A
Reference	(1) CIR Expert Panel. Safety assessment of alkyl glyceryl ethers as used in cosmetics. IJT 32(Suppl. 3):5-21, 2013.
POD (NOAEL)	50, 13 weeks

INCI	Prunus Amygdalus Dulcis Oil
Presence in the annexes	None
Description	Sweet Almond Oil is the fixed oil obtained from the ripe seed kernels of various species of <i>Prunus</i> . Almond kernels contain as much as 50% oil, which can either be physically expressed (35% yield) or chemically extracted (50% yield). Sweet Almond Oil consists mostly of triglycerides of oleic and linoleic acids. Oils derived from edible vegetables, fruits, seeds, tree, and ground nuts have been safely consumed by, and applied to the skin of, humans for thousands of years. The group of ingredients characterized as fats and oils are the glyceryl esters of fatty acids (triglycerides) normally found in plants. The raw oil may include diglycerides, monoglycerides, free fatty acids, plant sterols, pigments, glucosides, proteins, natural antioxidants, vitamins, and impurities. exposure to the oils from food use would result in a much larger systemic dose than that resulting from use in cosmetic products. <i>Cocos nucifera</i> (coconut) oil, oil from the dried coconut fruit, is composed of 90% saturated triglycerides. The primary constituents of coconut oil are trimyristin, trilaurin, tripalmitin, tristearin, and various other triglycerides. ¹¹ About 90% of the triglycerides are saturated.
Oral toxicokinetics	Sweet Almond Oil was well-digested (97%) when compared to other oils. In another study, Sweet Almond Oil or whole almonds were added to a lipid-free basal diet at concentrations of 10% and 18%, respectively, and fed to groups of three rats for three days. Absorption of the Almond Oil was 98%-99% complete.
Inhalation toxicokinetics	N/A
Dermal absorption	The percutaneous absorption of Sweet Almond Oil was studied in adult albino rats. A solution containing this ingredient and physostigmine was applied to the skin of each animal; latency time to muscle contraction was then measured. When compared to other oil mixtures tested, Sweet Almond Oil-physostigmine had a relatively long latency time, indicating lower percutaneous absorption.
Skin irritation	Undiluted Sweet Almond Oil was tested for irritancy in groups of six male albino rabbits. The test material was applied under occlusion to the clipped intact and abraded dorsal skin of each animal. Twenty-three hours later, patches were removed; sites were scored at 24 and 48 hours. The Primary

	Irritation Indices (PIIs) for seven test samples of Sweet Almond Oil ranged from 0 to 0.18 (maximum score = 8), indicating that this ingredient is practically nonirritating to skin.'
Eye irritation	slightly irritating
Skin sensitisation	not sensitising
Acute toxicity	Undiluted Sweet Almond Oil and products containing 2%-18% Sweet Almond Oil were tested for acute oral toxicity. Results of these tests indicate that undiluted Sweet Almond Oil and products containing up to 18% Sweet Almond Oil are practically nontoxic
Repeated dose studies	N/A
Mutagenicity/genotoxicity	Bacterial Reverse Mutation Assay of Almond Sweet Extract (CAS 90320-37-9) - test item does not show any mutagenic nor pro-mutagenic activity.
Carcinogenicity	N/A
Reproductive toxicity	N/A
Photosensitivity	Results of photo patch tests of formulations containing up to 2% Sweet Almond Oil in 764 subjects indicated that these products do not manifest photosensitization.
Endocrine effects	N/A
Reference	1. CIR Expert Panel Safety Assessment of Prunus Amygdalus Dulcis (Sweet Almond) Meal Seed 2. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/20014/7/11/6 as Used in Cosmetics, https://www.cir-safety.org/sites/default/files/RR_Sweet%20Almond.pdf
POD (NOAEL)	N/A

INCI	Carbomer
Presence in the annexes	None
Description	Carbomer is manufactured by reflux polymerization of acrylic acid in an inert solvent in the presence of a catalyst. In doing this, a closed system, free of oxygen and water, is used. It is a large molecule and remains in particulate form (dispersed) in final preparations and thus will not likely cross the stratum corneum to induce systemic toxicity.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	Does not penetrate
Skin irritation	low potential for irritation
Eye irritation	zero to moderate
Skin sensitisation	low potential for sensitisation
Acute toxicity	Acute oral studies with rats, guinea pigs, mice, and dogs showed that Carbomer has low toxicity when ingested. The inhalation LC50 of Carbomer in albino rats was 1.71 mg/l. The dermal LD50 of rats exposed to Carbomer was > 3 g/kg.
Repeated dose studies	Feeding of rats with doses up to 5.0 g/kg/day Carbomer (49 days) and of rats and dogs with up to 5.0% Carbomer in the diet (21 days) resulted in lower than normal body weights. In rats fed Carbomer at dietary levels of 5.0% for 90 days, absolute liver weights and liver to body and brain weight ratios were reduced, but no pathological changes were observed. Rats fed Carbomer at dietary levels of 0.1%, 0.5%, or 5.0% for 6.5 months exhibited various organ weight changes. In dogs fed 0.5 or 1.0 g/kg/day Carbomer for 6.5 months, gastrointestinal irritation

	and marked pigment deposition within Kupffer cells of the liver were observed. In another study, there were no significant effects in dogs fed up to 1.0 g/kg/day Carbomer for 32 months.
Mutagenicity/genotoxicity	N/A
Carcinogenicity	N/A
Reproductive toxicity	N/A
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. CIR Expert Panel Amended Safety Assessment of Acrylates Copolymers as Used in Cosmetics
POD (NOAEL)	N/A

INCI	Panthenol
Presence in the annexes	0
Description	Precursor of vitamin B5 (pantothenic acid), provitamin B5. Panthenol after penetration into the skin is converted into pantothenic acid. Both compounds have the same biological activities. Vitamin B is found in every living cell.
Oral toxicokinetics	With a molecular weight of 205.25 g/mol, a logKow value of -1.06 and a water solubility of >509 g/L D- Panthenol is likely to be absorbed in the GI tract. extensive gastrointestinal absorption is expected for D- Panthenol based on physical chemical properties.
Inhalation toxicokinetics	Based on the low vapour pressure of 0.000056 Pa D- Panthenol does not vaporise in a sufficient manner to become available for inhalation exposure. Exposure to aerosols may occur. However, due to the high water solubility the substance is likely to solve in the mucus lining of the respiratory tract but subsequent systemic absorption is not likely.
Dermal absorption	The substance characteristics and physical-chemical properties indicate that dermal absorption is likely. The physical state favours quick dermal absorption with liquids taken up more readily than dry particulates.
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not a dermal sensitizer
Acute toxicity	Oral: The LD50 was > 10000 mg/kg bw. No death were observed during a 7 h-inhalation exposure of rats to a saturated atmosphere of the test item at room temperature. Dermal: oral: The LD50 was > 10000 mg/kg bw. No death were observed during a 7 h-inhalation exposure of rats to a saturated atmosphere of the test item at room temperature.
Repeated dose studies	13 weeks, rat, The subchronic toxicity following voluntary consumption of the test item by male and female rats in drinking water in the concentrations of 200, 50 and 20 mg/kg bw/day for a 90 day period showed essentially negative results. The test item was concluded to be relatively innocuous under the conditions of the experiment. Signs of a mild eosinophilia appeared to be present among the experimental animals, but this has been seriously questioned. The no observed adverse effect level (NOAEL) under the conditions of this study was considered to be 200 mg/kg bw/day. NOAEL is the highest tested dose, but no adverse effects was reported; A read across approach was performed with the supporting substance DL-Ethylpanthenol. The repeated dose toxicity of DL-Ethylpanthenol was investigated in a subacute 28-day GLP and guideline oral toxicity study by daily gavage in the rat, the dose levels for the 28-day toxicity study were selected to be 0, 50, 200 and 1000 mg/kg bw/day. From the results presented in this report a definitive No Observed Effect Level (NOEL) of 1000 mg/kg bw/day was established.

Mutagenicity/genotoxicity	D,L-Panthenol was not mutagenic in the bacterial reverse mutation test using bacteria.
Carcinogenicity	N/A
Reproductive toxicity	There were no test article-related clinical findings or effects on maternal body weight, body weight gains, or food consumption observed at any dosage level. In addition, there were no test article-related effects on embryo/fetal development at any dosage level. Based on the results of this study, a dosage level of 1000 mg/kg/day, the highest dosage level evaluated, was considered to be the no-observed-adverse-effect level (NOAEL) for maternal toxicity and embryo/fetal development when DL- Ethyl Panthenol was administered orally by gavage to bred CrI:CD(SD) rats.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/14227/7/9/3
POD (NOAEL)	1000 mg&kg bw/day from 28 days study

INCI	Urea
Presence in the annexes	None
Description	Urea is produced in the body (liver) of mammals as a consequence of normal physiological processes, primarily by the detoxification of ammonia resulting from protein catabolism, via the urea cycle.
Oral toxicokinetics	The molecular weight below 500g/mol, the absence of ionisable groups, the hydrophilic properties and the particle size, are in favor of oral absorption. When SC (subcutaneous) versus oral LD50 values were compared, a factor of 0.57-0.63 was reached. Based on these data, an oral absorption of 60% is considered
Inhalation toxicokinetics	Based upon the solid state with high particle size (not reaching the thoracic airways), low vapor pressure and high water solubility, deposition in the lower airways is assumed most unlikely and absorption by inhalation is considered to be negligible. From a conservative viewpoint, a 10 % inhalation absorption is considered.
Dermal absorption	Based upon the low molecular weight, solid state and low vapor pressure (not evaporating), and the fact that the structure is not binding to skin molecules, dermal absorption is possible. However, the high water solubility, log P <0 and absence of irritation are not in favour of dermal absorption. Experimental data on the absorption of urea across normal human skin in vitro is reported to be up to 9.5%. From a realistic conservative viewpoint, a 10 % dermal absorption is considered.
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	LD 50 14 300 mg/kg bw
Repeated dose studies	In 12 -month carcinogenicity screening assays (Fleischman et al, 1980), F-344 rats and C57BL/6 mice (50/sex/group) were exposed to urea in the diet at concentrations of 4500, 9000 or 45000 ppm for 12 months. There were no signs of toxicity. Survival and bodyweights were unaffected by treatment. Gross and microscopic pathology did not reveal any treatment-related effects. It is concluded that urea is of very low chronic toxicity. Using default conversion factors, the dose level of 45000 ppm is calculated to be equivalent

	to approximately 2250 mg/kg bw/d in the rat and 6750 mg/kg bw/d in the mouse.
Mutagenicity/genotoxicity	No classification is proposed for genotoxicity. Urea is produced by the body in large quantities as a normal product of metabolism and is present in the bloodstream at high concentrations. Urea is therefore considered extremely unlikely to be genotoxic.
Carcinogenicity	No classification is proposed for carcinogenicity. There is no evidence from animal studies that urea is carcinogenic. The physiological role of urea and level of production by the human body indicates that the substance is not carcinogenic.
Reproductive toxicity	No standard studies are available. It is considered extremely unlikely that occupational, primary or secondary exposure to urea will result in any effects on fertility as the levels of exposure will be insignificant compared to those present in the body as a result of protein catabolism.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/16152/7/9/1
POD (NOAEL)	2250mg/kg bw/day, 12 months, rat

INCI	Sodium Benzoate
Presence in the annexes	V/1
Description	Present in Annex V entry 1 of 1223/2009 Directive as a preservative, concentration limited as acid to 0.5% in leave-on products, 2.5% in rinse-off products except oral products and 1.7% in oral products. In the United States, benzoic acid and sodium benzoate are on the FDA list of substances that are generally recognized as safe (GRAS). Both may be used as antimicrobial agents, flavouring agents and as adjuvants with a current maximum level of 0.1% in food. Sodium Benzoate, expressed as benzoic acid, is permitted in oral medicines up to 0.5%, in parenterally administered up to 0.5%.
Oral toxicokinetics	Benzoic acid and its sodium and potassium salt are considered with benzylacetate, benzylalcohol, benzaldehyde as a single category from the human health view by JECFA, as they are all rapidly metabolized and excreted via a common pathway within 24hrs. Sodium benzoate is expected to immediately dissociate and form benzoic acid in an aqueous environment.
Inhalation toxicokinetics	0
Dermal absorption	0
Skin irritation	Application of 50 mg sodium benzoate/rabbit for 24 h; responses were scored at 24 h, 48 h and 72 h; post-exposure observation time: 7 d. Results: not irritating.
Eye irritation	0
Skin sensitisation	There were 2 additional human patch test references that were not included with any of the submissions. The first was a large cohort of 2045 patients, 5 gave positive results. These gave occasional positive results (Brasch, J. et al., 1993). In the second test, patch tests gave positive nonimmunologic contact urticaria (Nethercott, J.R., 1984).
Acute toxicity	LD 50 1714
Repeated dose studies	Groups of 4-5 male and 4-5 female rats received 0, 1, 2, 4, and 8 % sodium benzoate equivalent to 640, 1320, 2620, and 6290 mg/kg bw/d in the diet for 90 days. 4/8 animals died (average 13 days to death) in the 8 % dose level group, the average weight gain of the surviving rats was reduced and the relative liver and kidney weight was significantly increased with histopathological changes in liver

	and kidney (7/16). Data are insufficient to justify a NOAEL of 4 % in the diet (2.6 g/kg bw/d).
Mutagenicity/genotoxicity	Sodium benzoate (up to 3.0 mg/plate) was tested in the Salmonella/microsome test using <i>S. typhimurium</i> TA 92, TA 94, TA 98, TA 100, TA 1535 and TA 1537. No significant increases in the numbers of revertant colonies were detected in any <i>S. typhimurium</i> strains at the maximum dose. Chromosome aberration test was carried out on sodium benzoate using human embryonic lung culture cells. Sodium benzoate produced no significant increase in the aberration frequency in the anaphase chromosomes when tested at the dosage levels 0, 2.0 µg/ml, 20 µg/ml and 200 µg/ml.
Carcinogenicity	Sodium benzoate was given in 2 % in drinking water to 50 female and 50 male Swiss albino mice from weeks 5 on for lifespan. The average daily intake of sodium benzoate was 119.2 mg for a female and 124.0 mg for a male (approx. 5.95 - 6.2 g/kg bw/d). There was no effect of the survival of the treated mice when compared with the untreated control. There were no significant differences between the tumour distribution in sodium benzoate-treated and untreated control mice.
Reproductive toxicity	Sodium benzoate at doses of 0, 1.75, 8, 38 and 175 mg/kg bw was administered by gavage to groups of at least 20 pregnant albino CD outbred mice and White albino rats on gestation day 6 to 15. The NOEL in these studies was in all species identical with the highest dose tested:
Photosensitivity	In the JECFA report, there was a summary of an in vitro test by Eberlein-König et al, 1993. Suspensions of human erythrocytes were incubated with Benzoic Acid and Sodium Benzoate. Each material was tested at 10 ⁻⁵ , 10 ⁻⁴ and 10 ⁻³ mol/l. Erythrocyte-free samples were also incubated with the test material and used as controls. Following incubation, suspensions and samples were exposed to varying amounts of UVA light from one of three sources. Hemolysis was measured as a function of absorbance of 550 nm light. The substances did not produce significant photohaemolysis.
Endocrine effects	N/A
Reference	1. SCCP Opinion on Benzoic Acid https://ec.europa.eu/health/ph_risk/committees/04_sccp/docs/sccp_o_015.pdf
POD (NOAEL)	500mg/kg bw day from a 4 generation study

INCI	Triethanolamine
Presence in the annexes	None
Description	TEA is an amino alcohol commercially produced by aminating ethylene oxide with ammonia. CIR Expert Panel: Safe for use in cosmetic formulations designed for discontinuous, brief use followed by thorough rinsing from the surface of the skin; in products intended for prolonged contact with the skin, the concentration of diethanolamine should not exceed 5%; should not be used with products containing N-nitrosating agents. The TEA contains small amounts of diethanolamine and ethanolamine. The TEA is reactive and bifunctional, combining the properties of alcohols and amines. The reaction of ethanolamines and sulfuric acid produces sulfates. The TEA can act as an antioxidant against the autoxidation of fats of both animal and vegetable origin. Of concern in cosmetics is the conversion (N-nitrosation) of secondary amines (R1

	-NH-R2 , such as diethanolamine (wherein R1 and R2 are each ethanol), into N-nitrosamines that may be carcinogenic. Tertiary alkyl amines (NR1 R2 R3), such as TEA (wherein R1, R2, and R3 are each ethanol), however, do not tend to react with N-nitrosating agents to directly form nitrosamines. However, tertiary amines can act as precursors in nitrosamine formation by undergoing nitrosative cleavage (eg, 1 ethanol functional group can be cleaved off from TEA to generate diethanolamine). The resultant secondary amine (ie, diethanolamine) can then be N-nitrosated (ie, to N nitrosodiethanolamine [NDELA]). Accordingly, TEA can react, in a formulation or in vivo, with nitrites or oxides of nitrogen to form a nitrosamine. Nitrous anhydride is the oxide of nitrogen that most commonly initiates nitrosation in vivo.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	Coupled with the small actual exposure in the breathing zone and the concentrations at which the ingredients are used, this information suggested that inhalation would not be a significant route of exposure that might lead to local respiratory or systemic toxic effects.
Dermal absorption	Studies in experimental animals indicated that TEA is absorbed through the skin. In mice, most of the topically applied 14C-TEA is absorbed, and only 2% to 11% is detected at the site of application after 48 hours (Dow 1988,1989; Stott, 2000). The dermal absorption of TEA in rats was less extensive and much slower than in mice (Dow, 1988,1989).
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	not sensitising
Acute toxicity	Acute toxicity data indicate low toxicity: in rats the oral LD50 was 6400 mg/kg bw; in rabbits the dermal LD50 was > 2000 mg/kg bw. Inhalation exposure for 8 hours to vapour saturated with TEA failed to cause any deaths in rats (LC50 was not determined).
Repeated dose studies	Oral: In a sub-chronic oral toxicity study, a NOAEL of 1000 mg/kg bw/day was established, the highest dose tested. Dermal (90 days): In a sub-chronic dermal toxicity study, NOAELs of 125 and 250 mg/kg bw/day were established for local effects for males and females. Systemic NOAELs of 125 and 250 mg/kg bw/day were determined for males and females, respectively, based on kidney effects.
Mutagenicity/genotoxicity	TEA did not cause gene mutations in Salmonella typhimurium and Escherichia coli (Ames test), nor were chromosomal aberrations or sister chromatid exchanges induced in Chinese hamster ovary cells observed. An in vitro gene mutation assay (mouse lymphoma (L5178Y TK+/-) forward gene mutation assay) was also negative. All tests were performed in the absence and presence of metabolic activation.
Carcinogenicity	n increase in tumor incidence was neither observed in rats treated dermally or orally with TEA for 2 years, nor in mice exposed to TEA by oral administration for 2 years. In a dermal mouse carcinogenicity study there was an increased incidence of liver tumors in females at all doses tested. Mechanistic research specifically on TEA indicates that, to the extent TEA can potentially induce tumors in mice, it does so by a mechanism that is not relevant to humans. Therefore, based on the available data, TEA is not considered carcinogenic for humans.

Reproductive toxicity	No two-generation reproduction toxicity study is available for TEA. Under the conditions of a two-generation reproduction toxicity study with MEA HCl, the NOAEL for systemic toxicity and fertility, reproductive performance in parental F0 and F1 Wistar rats is 300 mg/kg bw/day. The NOAEL for pre-and postnatal developmental toxicity in their offspring is 1000 mg/kg bw/day.
Photosensitivity	TEA at cosmetic use concentrations was not phototoxic or photoallergenic in non-human and human studies, and TEA-lauryl sulfate was not photoallergenic in human studies.
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/15134/7/9/1 2. CIR Expert Panel Safety Assessment of Triethanolamine and Triethanolamine-Containing Ingredients as Used in Cosmetics
POD (NOAEL)	300 mg/kg bw/day, repr. Toxic. 2 generations, oral unknown

INCI	Achillea Millefolium Extract
Presence in the annexes	None
Description	Achillea millefolium is an herbaceous plant with characteristic narrow, oblong, multiple pinnate leaves. A sample of an achillea millefolium extract (aqueous) mixture (water 73.5%, butylene glycol 20%, pentylene glycol 5%, achillea millefolium extract 1%, xanthan gum 0.5%) contained 3.37% polyphenols, 61.25% proteins, and 38.12% sugars. Achillea millefolium is reported to contain linalool (1-4,000 ppm), thujone, quercetin, a-peroxyachifolid, and hydroquinone. The CIR Panel concluded that achillea millefolium extract, achillea millefolium flower extract (Note 1), and achillea millefolium flower/leaf/stem extract (Note 1) are safe in cosmetics when formulated to be nonsensitizing.
Oral toxicokinetics	No new toxicokinetics data were identified or made available for review. However, since botanical extracts are mixtures, toxicokinetic data would only have meaning for individual constituents of the extract, not for the extract as a whole.
Inhalation toxicokinetics	There were no inhalation toxicity data available. However, the Panel believes that the sizes of a substantial majority of the particles of the products containing these ingredients, as manufactured, are larger than the respirable range and/or aggregate and agglomerate to form much larger particles in formulation.
Dermal absorption	N/A
Skin irritation	not irritating
Eye irritation	not irritating
Skin sensitisation	Achillea millefolium is a member of the Asteraceae (formerly Compositae) family, which is known to be sensitising. An aqueous A millefolium extract was not a sensitizer in an LLNA at 1%. Two products containing Achillea millefolium extract up to 0.001133% were not sensitizing in HRIPTs. A product containing achillea millefolium extract at 0.04% was neither irritating nor sensitizing. A woman was reported to develop an allergic reaction to A millefolium after working with dried flowers.
Acute toxicity	Acute dermal toxicity data on A millefolium (yarrow)-derived ingredients were not found in the published literature, and no unpublished data were

	submitted. The oral LD50 of the mixture containing an aqueous extract of A millefolium was reported to be >2,000 mg/kg in female rats.
Repeated dose studies	An aqueous A millefolium extract (0.3, 0.6, 1.2 g/kg/d; leaves, stalks, stems) orally administered to male and female Wistar rats (n ¼ 10/sex) for 28 or 90 consecutive days produced no signs of toxicity. The authors concluded that the rats had no treatment-related toxicological or histopathological abnormalities. An ethanol (60%) multiherb mixture (20 mg/kg/d) that included achillea millefolium extract (3.5%; 0.7 mg/kg), orally administered to CBA/HZb mice (n ¼ 6) for up to 6 months, caused no clinical signs.
Mutagenicity/genotoxicity	In an Ames test using Salmonella typhimurium (TA98, TA100, TA102, TA1535, TA1537), the mixture containing an aqueous extract of A millefolium (0.06-5 mL) was not mutagenic with or without metabolic activation. In 2 micronucleus tests using V79 cells, the mixture containing an aqueous extract of A millefolium (up to 15 000 mg/mL) was not clastogenic or aneugenic with or without metabolic activation. 6 In a gene mutation assay using mouse lymphoma L5178Y TKp/, a product (up to 5 mL/mL) that contained an aqueous extract of A millefolium (0.5%) was not mutagenic with or without metabolic activation.
Carcinogenicity	N/A
Reproductive toxicity	An ethanol (45%) A millefolium extract (2.8 g/kg/d; 56 times the equivalent of a human dose of 50 mg/kg/d) was not maternotoxic when orally administered to Sprague Dawley rats (n ¼ 5) but caused decreased body weights in fetuses. An aqueous A millefolium leaf extract (0.3, 0.6, 1.2 g/kg/d) orally administered to male Wistar rats (n ¼ 10) for 90 days was not toxic nor caused any clinical or behavioral signs, but there was an increase in abnormal sperm in the males in the high-dose group. An ethanolic A millefolium flower extract (200, 400, 800 mg/kg) IP or orally administered to male albino Wistar rats (n ¼ 5) every other day for 22 days caused no changes in the low-dose IP group and the low- and mid-dose oral groups; however, there were abnormalities in the development of sperm in the mid- and high-dose groups. Although A millefolium-derived ingredients caused an increase in abnormal sperm and damage to male organs in rats, these effects were observed at levels much greater than any from exposure to cosmetics.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. CIR Expert Panel Safety Assessment of Achillea millefolium as Used in Cosmetic, https://online.personalcarecouncil.org/ctfa-static/online/lists/cir-pdfs/PRS627.pdf
POD (NOAEL)	400mg/kg bw/day from reprod. Toxicity test 20 days

INCI	Hypericum Perforatum Extract
Presence in the annexes	None
Description	A common name for this plant is St John wort. Hypericum perforatum extract is an extract of the capsules, flowers, leaves, and stem heads of the Hypericum, H perforatum. In 1998, it was reported to the Food and Drug Administration (FDA) that H perforatum extract and H perforatum oil were used in 64 and 11 cosmetic formulations,

	respectively. <i>Hypericum perforatum</i> flower contains not less than 0.08% of total hypericins expressed as hypericin calculated with reference to the dried drug. Constituents of <i>H. perforatum</i> include: phloroglucinol derivatives: 0.2% to 4%, depending on the age of the herbal drug, mainly hyperforin and its homolog adhyperforin and furanohyperforin; naphthodianthrone: 0.06% to 0.4%, mainly pseudohypericin and hypericin, protohypericin, protopseudohypericin, cyclopseudohypericin, and skyrin derivatives; the amount of pseudohypericin is about 2 to 4 times higher than that of hypericin; flavonoids: 2% to 4%, mainly glycosides of the flavonol quercetin: hyperoside, rutin, isoquercitrin, quercitrin; also biflavones (I3, I18-biapiogenin, amentoflavone); procyanidines: for example, procyanidine B2, tannins with catechin skeletal (6%-15%); xanthones: in trace amounts, essential oil: 0.1% to 0.25%; the essential oil of dried flowering tops contains as main compounds 2-methyloctane (16%) and α-pinene (10.6%). In the essential oil of leaves of Indian origin, 58 components were identified, α-pinene (67%) being dominant; the other components included caryophyllene, geranyl acetate, and nonane (each about 5%); other constituents: include small amounts of chlorogenic acid and other caffeoylquinic and p-coumaroylquinic acids and also free amino acids. In a batch of <i>St. John wort</i> extract capsules, the label stated that they contained 300 mg of extract and 900 mg of hypericin.
Oral toxicokinetics	After a single oral dose of <i>H. perforatum</i> extract (300 mg; tablet form; 900 mg hypericin $\pm$ pseudohypericin), the mean serum level in subjects ($n = 12$) of total hypericin $\pm$ pseudohypericin was 43 ng/mL. The authors stated that these skin levels are far below hypericin skin levels that are estimated to be phototoxic (>100 ng/mL). Repeated doses of <i>H. perforatum</i> extract (1600 mg/kg/d for 8 days) caused a continuous increase in the plasma levels of quercetin and isorhamnetin for 5 days; after that time the concentration remained constant.
Inhalation toxicokinetics	N/A
Dermal absorption	The distribution of hypericin-related fluorescence in the skin after dermal administration (1%) was concentrated in the stratum corneum and epidermis with only faint fluorescence in the dermis observed. At lower concentrations (0.1% and 0.01%), the fluorescence was concentrated only in the stratum corneum and was faint in the epidermis.
Skin irritation	A mixture of <i>H. perforatum</i> oil, butylene glycol, and water was not irritating in clinical studies. In an irritation test ($n = 18$), a bath oil containing <i>H. perforatum</i> extract (concentration not provided; 50 mL) did not cause irritation and was similar to the control of distilled water.
Eye irritation	<i>Hypericum perforatum</i> may increase the risk of cataracts but the mechanism is not established. Human lens epithelial cells incubated in hypericin (0.1-10 mmol/L) and irradiated (4 J/cm² UVA or 0.9 J/cm² visible light) had increased necrosis and apoptosis.
Skin sensitisation	N/A
Acute toxicity	Intravenous administration of hypericin (2 mg/kg in 2% benzyl alcohol and saline) was well tolerated by rhesus monkeys ($n = 3$).²⁵ At a dose of 5 mg/kg, a transient severe photosensitivity rash was observed at 12 hours that resolved within 12 days. Edema and a pruritic erythematous rash with evolution to eschar were observed on the face and lightexposed skin. Mild anorexia and transient elevation in hepatic transaminases were observed.
Repeated dose studies	Oral therapeutic use <i>H. perforatum</i> was reported to be safe up to 900 mg/d (≈ 13 mg/kg/d) for humans.
Mutagenicity/genotoxicity	In an Ames test, a tincture of <i>Hypericum</i> had mutagenic effects at 20 mg/100 mL suspension, which the researchers attributed to flavonols. However, the

	origin of the plant and the mode of preparation of the tincture were considered to play a role in the mutagenic potential. In another Ames test, H perforatum (10 mL) had mutagenic activity; in testing fractions of 3 extracts, the mutagenic potential was found exclusively in quercetin, and hypericin was not mutagenic. Hypericum extract (500 mL) and hypericin were not genotoxic in unscheduled DNA synthesis assays using primary rat hepatocytes. Hypericum extract (4.00 mL/mg) was not mutagenic in a cell transformation assay using Syrian golden hamster embryo cells, and it was not genotoxic in a mouse fur spot test or in a chromosome aberration test. Quercetin may be genotoxic and is reported to be in H perforatum plant parts at 1000 to 20 000 ppm. Because the maximum concentration of use in cosmetics that contain these H perforatum extracts was reported to be 0.07%, the Panel concluded that the amount of exposure to these constituents would be below the threshold of toxicological concern.
Carcinogenicity	N/A
Reproductive toxicity	There were no reproductive or developmental effects observed in a 2-generational study of H perforatum extract using CD-1 mice (n ¼ 20). There were no clinical signs of maternal or developmental toxicity when pregnant Wistar rats (n ¼ 15) were orally administered H perforatum extract (36 mg/kg/d in saline; 0.4% hypericin) during gestation days 9 to 15. In 36 samples of breast milk from mothers (n ¼ 5) who were taking H perforatum extract (300 mg times/d), hyperforin was present in the milk at 0.9% to 2.5%. 21 The plasma from 2 of the infants contained low levels of hyperforin (0.1 ng/mL). No side effects were seen in the mothers or infants. The authors concluded that these results add to the evidence of the relative safety of St John wort while breastfeeding. Hyperforin was detected in the breast milk of a mother took 3 Hypericum extract pills (3 300 mg/d; 0.12%-0.28% Hypericum, *4.5% hyperforin).22 No clinical effects were observed in the mother and infant.
Photosensitivity	A product containing H perforatum extract (1.1%) was not photosensitizing to the backs of guinea pigs when applied to tape-stripped skin.
Endocrine effects	N/A
Reference	1. CIR Expert Panel Amended Safety Assessment of Hypericum Perforatum-Derived Ingredients as Used in Cosmetics
POD (NOAEL)	13mg/kg bw/day, duration unknown, NOAEL from humandata

INCI	Tocopheryl Acetate
Presence in the annexes	0
Description	Tocopheryl acetate is in terms of Regulation (EC) No 1907/2006 (REACH) exempt from the obligation to register, as sufficient information is known about this substance that it is considered to cause minimum risk because of its intrinsic properties. SCCNFP is of the opinion that alpha-tocopherol acetate does not pose a threat to the health of the consumer and therefore does not propose any restrictions or conditions on the use of alpha-tocopherol acetate in cosmetic products. The safety of Tocopheryl Acetate has also been assessed by the CIR Expert Panel. The CIR Expert Panel evaluated the scientific data and

	concluded that this ingredient is safe as used in cosmetics at concentrations up to 36% (100% in Vitamin E oil).
Oral toxicokinetics	In a dermal absorption study using human subjects, tocopheryl acetate was substantially absorbed in the skin, but systemic availability was not observed. Also, conversion to tocopherol was not seen. In a study using rats, approximately 6% of the applied dose penetrated into the epidermis after 5 days. Most studies found that some tocopheryl acetate was converted to tocopherol. Irradiation of the animals dosed with tocopheryl acetate resulted in a significant increase in the amount of tocopheryl acetate found in the skin as compared to non-irradiated animals, and irradiation significantly increased the amount of tocopherol found in the skin of test animals.
Inhalation toxicokinetics	N/A
Dermal absorption	Dermally applied tocopheryl acetate is hydrolyzed to tocopherol upon exposure to UV. In a study using rats, approximately 6% of the applied dose penetrated into the epidermis after 5 days.
Skin irritation	Not irritating (rabbit)
Eye irritation	Not irritating to moderately irritating (undiluted, rabbit)
Skin sensitisation	Not sensitizing (undiluted, human). It produced skin sensitization in one animal test, was not sensitizing in the "guinea pig maximization test", but can cause sensitization in the "open epicutaneous test" (>30% tocopheryl acetate).
Acute toxicity	Oral LD50 (rat) > 16 g/kg; Oral LD50 (mouse) > 4 g/kg
Repeated dose studies	Tocopheryl acetate was not toxic in an 8-wk feeding study. A short-term repeated dose toxicity study (28 days) resulted in the following NOAEL values (racemic tocopheryl acetate): Rat: 1111 mg/kg; Dog: 360 mg/kg.
Mutagenicity/genotoxicity	Not genotoxic in Ames tests or a chromosomal aberration assay
Carcinogenicity	Tocopheryl acetate (≤ 2 g/kg/day) was not carcinogenic in a dietary study. Dermally applied tocopheryl acetate was reported to enhance photocarcinogenesis. Orally administered tocopheryl acetate reduced the incidence of skin cancer, but toxicity was observed.
Reproductive toxicity	Oral administration of tocopheryl acetate (≤ 1.6 g/kg/day) generally did not have any reproductive or developmental effects in rabbits, hamsters, rats, or mice. Tocopheryl acetate had some effect on reducing the number of malformations observed in neonates from diabetic dams. In some studies, tocopheryl acetate potentiated the embryo-lethal effects of cortisone acetate.
Photosensitivity	Not photoallergenic (guinea pig); photoprotective effects observed in mice and humans
Endocrine effects	N/A
Reference	(1) Norwegian Food Safety Authority. Risk profile Vitamin E. 10.12.2012. (NOAEL) (2) CIR Expert Panel. Safety Assessment of Tocopherols and Tocotrienols as Used in Cosmetics. April 4, 2014. (3) Dweck A. C. Handbook of Cosmetics Ingredients - Their Use, Safety and Toxicology. Third edition, 2012.
POD (NOAEL)	360, 28days

INCI	Retinyl Palmitate
Presence in the annexes	None
Description	Vitamin A is a micronutrient essential for most mammalian species and is one of three fat soluble Vitamins. The term "Vitamin A" is used as a generic descriptor for compounds that exhibit the biological properties of retinol or other closely related naturally occurring derivatives. Vitamin A occurs in

	nature primarily as retinyl ester and retinol. It is an essential factor for the growth and maintenance of higher organisms. It is required for visual function, epithelial cell differentiation and reproduction.
Oral toxicokinetics	Dietary Vitamin A, which is mostly in the form of retinyl esters, is absorbed in the upper part of the small intestine by mechanisms similar to those for lipid absorption. The esters undergo hydrolysis to release retinol, which is incorporated into mixed micelles and absorbed by enterocytes, where it is bound to an intra-cellular protein called CRBP (cellular retinol binding protein II). It is then re-esterified to form retinyl esters. The esters are incorporated into chylomicrons and are hydrolyzed in the general circulation. Chylomicron remnants are taken up by tissues, particularly the liver. Remnants are degraded within the hepatocytes, and the released retinol is transferred to stellate cells for storage after re-esterification. In the present study on basic toxicokinetics, the exposure to retinyl palmitate and its main metabolites in plasma was examined within the framework of prenatal developmental toxicity studies in cynomolgus monkeys (Hendrickx 2000). Retinyl palmitate was administered daily to female monkeys by nasogastric intubation at dose levels of approx. 4.1, 11, 22, 44 mg/kg (7500, 20000, 40000, 80000 IU/kg) during early pregnancy (i. e. between gestational days 16 and 27). Control monkeys were administered the vehicle (physiological saline or distilled water). Administration of retinyl palmitate resulted in the formation of the following metabolites: retinol, retinyl esters, all-trans-retinoic acid, 13-cis-retinoic acid, all-trans-4-oxo-retinoic acid, 13-cis-4-oxo-retinoic acid. In a weight of evidence, the dermal absorption rate of retinol and the respective esters in humans is considered to be approx. 5% of the applied dose. Considering the comparative assessment of oral versus dermal administration observed by Nohynek et al. (Nohynek 2006), the oral absorption is expected to be extensively higher (up to 15 fold increases), which substantiates the assumption of a 100% oral absorption of retinyl palmitate.
Inhalation toxicokinetics	In light of the present data from acute oral toxicity studies, and the fact, that inhalative exposure is not considered to be the major route of exposure, an acute toxicity study via the inhalative route is scientifically not required and is not in line with animal welfare requirements.
Dermal absorption	In a weight of evidence, the dermal absorption rate of retinol and the respective esters in humans is considered to be approx. 5% of the applied dose.
Skin irritation	Although the present key study on retinyl palmitate does not meet the criteria for classification as a skin irritant according to directive 67/548/EEC, the supportive study and the findings from structurally similar substances, i.e retinyl acetate and retinyl propionate, provide evidence for dermal irritative potential of this substance group, warranting a respective classification. However, the dermal irritative potential of retinyl palmitate appears to be lower than for retinyl acetate and retinyl propionate on the basis of the present data.
Eye irritation	The present data on eye irritation do not fulfill the criteria laid down in 67/548/EEC and regulation 1272/2008/EC, and therefore, a non-classification is warranted.
Skin sensitisation	not sensitising
Acute toxicity	The acute toxicity of retinyl-palmitate in animals is low, based on the data of structural analogue retinyl-propionate. The available acute oral toxicity study upon oral application of retinyl propionate (approx. 95% purity) was selected as the key study (performed according OECD 401 guideline, BASF 10A0454/891109) The oral LD50 was > 2000 mg/kg bw in rats in this well conducted limit study. In an acute oral toxicity study in rats, available as short summary from a secondary source, a LD50 = 7910 mg/kg has been given for retinyl-palmitate (Kamm 1987).

Repeated dose studies	A repeated dose 90 days (feed) study in rats was available on structural analogue retinyl acetate that was performed similar to OECD guideline 408: The NOAEL was 1.43 -3.07 mg/kg bw/d (28 ppm) based on slight effects on a few clinical chemical parameters (increased plasma triglyceride levels, slightly increased activity of glutamate pyruvate transaminase and glutamate oxalacetate transaminase) and effects in the liver (increased absolute and relative liver weights, brightened livers, lipid accumulation of the Kupffer cells and decrease of the intrahepatocellular lipid droplets in the peripheral areas of the liver lobules). According to the dose response assessment by the scientific committee on food (EFSA 2006) mainly on the basis of human surveys, the lowest doses reported to produce the major adverse effects were 7500 RE/day (representing approx 13.5 mg retinyl palmitate/ day) for bulging fontanelles, hepatotoxicity and lipid metabolism, > 3000 RE/day (representing approx 5.5 mg retinyl palmitate/day) for teratogenicity. Data concerning decreased bone density with focus on postmenopausal women at lower dose levels exist, but were considered to be insufficient to provide a sound evidence of causality between vitamin A intake and increased risk for bone fractures.
Mutagenicity/genotoxicity	On the basis of all available information on genotoxicity of retinyl palmitate and relevant studies on its respective structural and metabolic analogues retinol and retinyl acetate, retinyl palmitate is defined to be non-genotoxic in a weight of evidence.
Carcinogenicity	On the basis of all available information on genotoxicity of retinyl palmitate and relevant studies on its respective structural and metabolic analogues retinol and retinyl acetate, retinyl palmitate is defined to be non-genotoxic in a weight of evidence.
Reproductive toxicity	In this array of developmental toxicity studies in cynomolgus monkeys (<i>Macaca fascicularis</i>) retinyl palmitate was orally (nasogastric intubation) administered at doses of 4.1, 11, 22 and 44 mg/kg bw/d (7 500, 20 000, 40 000, 80 000 IU/kg bw/d, respectively) from GD 16 to GD 27 (Hendrickx 2000). Taken together, from the variety of available data on developmental toxicity and retinoids, the cynomolgus monkey was shown to be a sensitive animal species and the results obtained are considered as most relevant for human risk assessment. In the series of prenatal developmental toxicity studies in the pregnant Cynomolgus monkeys, oral doses of retinyl palmitate at 20000 and 40000 IU/kg bw/day resulted in developmental toxicity and maternal toxicity (Hypervitaminosis A), respectively, leading to an NOAELdevelopmental of 4.1 mg/kg bw/d and a NOAELmaternal of 11 mg/kg bw/d.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/13687/7/6/2
POD (NOAEL)	1.43 mg/kg bw/day from 90 days oral study

INCI	Tocopherol
Presence in the annexes	0
Description	Tocopherol and tocopheryl acetate are generally recognized as safe (GRAS) food ingredients. Systemic exposure resulting from cosmetic use is not expected to exceed the levels found safe for food use. Together, the tocopherols and tocotrienols comprise vitamin E
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A

Dermal absorption	The effect of vehicle on the dermal penetration of α -tocopherol was determined in micro-Yucatan pig skin. ⁴⁰ The dermal penetration of 1% tocopherol was evaluated in several vehicles, and α -tocopherol permeated into the receptor fluid regardless of the vehicle used. The greatest permeation found in viable skin + the receptor was from an emulsion vehicle that contained 10% isopropyl myristate; 12.24% of the applied dose was found in the viable skin and receptor fluid. The lowest penetration, 6.47% applied dose, was observed with an alcoholic vehicle that was 96% SD alcohol.
Skin irritation	non to mild irritating
Eye irritation	non to mild irritating
Skin sensitisation	N/A
Acute toxicity	In rats, the dermal LD50 is >3 g/kg. In mice, the oral LD50 of tocopherol is >25 ml/kg
Repeated dose studies	13 weeks study on rats, the NOAEL is set at 500 mg/kg.
Mutagenicity/genotoxicity	The test compound did not induce any relevant increase of the number of revertant colonies/plate in any of the five tester strains. Thus it can be concluded that all - rac a tocopheryl-acetate is not mutagenic in the Ames test under the described experimental conditions.
Carcinogenicity	N/A
Reproductive toxicity	No apparent differences in reproductive indices were seen between controls and treated groups of parents. Mean gestation period, litter size, sex ratio, and mortality of pups and parents were unaffected by the test substance. Hematology and clinical chemistry done after 255 days of treatment revealed no toxicologic differences between control and high dose group. None of the organ weights (neither absolute nor relative) of the treated groups were significantly different from control. Microscopic examination of tissues of high dose and control F0 and F1b animals revealed no morphological changes that were attributable to ingestion of the test substance. Ingestion of the test substance had no effect on body weight gain of the pups. Based on default values in the REACH guidance Chapter R.8: Characterisation of dose [concentration]-response for human health (Table R8 -17); using the daily intake of 20 gram/day; and the average body weight of the rats of 0.5 kg (data of the 90 day study), the NOAEL of 2% is converted to 800 mg/kg bw/day.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/sl/registration-dossier/-/registered-dossier/12703/7/9/2 2. CIR Safety Expert Panel https://www.cir-safety.org/sites/default/files/tocopherols.pdf
POD (NOAEL)	500 from 13 weeks study on rats

INCI	Eugenia Caryophyllus Leaf Oil
Presence in the annexes	None
Description	According to the European Pharmacopoeia (1/2008: 1091), Caryophylli flos consists of the whole flower buds of Syzygium aromaticum (L.) Merril et L.M. Perry (Syn. Eugenia caryophyllus (C. Spreng.) Bull. et Harr.) which were dried until they become reddish-brown. They contain not less than 150 ml/kg essential oil. Three main components account for nearly 99% of the essential oil: eugenol (75–88%), acetyl eugenol (4–15%), β -caryophyllene (5–14%). The medicinal use of Caryophylli floris aetheroleum can be traced in literature back to the 15th century. Clove essential oil acts in high concentrations as local irritant, allergic reactions may also be possible.

	However, when applied in diluted form, no reports on severe adverse events are published. When applied correctly in the proposed routes of administration, clove essential oil can be considered as clinically safe.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	In concentrated form, oil of clove may be irritating to mucosal tissues (Gruenwald et al 2004). In contrast to this Anton et al (2001) report that there is no skin irritation (undiluted oil) on hairless mice. Under occlusion the undiluted clove oil was moderately irritating in rabbits.
Eye irritation	0
Skin sensitisation	In a patch test study a 10% ethanol extract of Caryophylli flos was investigated among other herbal preparations used in the Traditional Chinese Medicine. Out of 30 patients 8 reacted positively to clove extract (Chen et al 2003).
Acute toxicity	LD50 1.8–3.72 g/kg
Repeated dose studies	Clove essential oil in oral dosages of 35 or 70 mg per animal (rat) over 8 weeks was tolerated without signs of toxicity. Higher doses led to inactivity and weight loss. 105 mg/animal p.o. daily for 2 to 3 weeks led to serious liver and kidney damage and death of the animals. Clove oil is allowed as food additive and therefore an administration to food-producing animals is possible. The allowed daily intake for the United States is reported with 2.5 mg/kg
Mutagenicity/genotoxicity	The dry residue of aqueous and methanolic extracts showed mutagenic effects in the rec assay in <i>Bacillus subtilis</i> . The mutagenic activity in the Ames test on <i>Salmonella typhimurium</i> TA98 and TA100 was not assessable due to the antimicrobial action. An in vivo bone marrow micronucleus assay demonstrated that the administration of 0.5% and 2% of cloves in the diet of mice for 10 days neither significantly induced micronuclei nor could effectively modulate the 7, 12-dimethylbenz[a]anthracene genotoxicity in mice
Carcinogenicity	Rats: eugenol was administered in the diet in a concentration of up to 6000 ppm. Considering the mean diet consumption per day the daily uptake of eugenol was for male rats about 1100 mg/kg body weight at the beginning to 245 mg/kg body weight at the end of the study; for female rats about 790 mg/kg body weight at the beginning to 237 mg/kg body weight at the end of the study. The study outcome was negative in both sexes.
Reproductive toxicity	Domaracky et al (2007) investigated the effects of clove essential oil on the growth and development of mouse pre-implantation embryos in vivo. The animals received 0.25% essential oil in the commercial diet (= 375 mg/kg per day). Treatment with clove essential oil induced a significantly higher percentage of dead cells compared to the control group. Female CD-1 mice were given clove oil from day 6 to day 15 of gestation p. o. 2.2 to 215 mg/kg body weight daily. The foetuses were obtained on day 17. The use of clove oil had no apparent toxic effect on the implantation and survival of the mother and foetus. The number of malformations of the soft tissues and the skeletal system was not different from that in the control group spontaneously occurring.
Photosensitivity	N/A
Endocrine effects	N/A

Reference	1. EMA Assessment report on Syzygium aromaticum (L.) Merrill et L.M. Perry, flos and Syzygium aromaticum (L.) Merrill et L.M. Perry, floris aetheroleum
POD (NOAEL)	2.5mg/kg bw/day ADI

INCI	CALENDULA OFFICINALIS FLOWER EXTRACT
Presence in the annexes	None
Description	The Cosmetic, Toiletary, and Fragrance Association (CTFA) provided a specification for pot marigold from Alban Muller International that included 12% glycosides (roughly 6% mucilages), 5.3% lipids, 10% minerals, and 0.02% essential oil with oxygenated sesquiterpenic derivatives, along with organic acids; phenolic compounds, including phenolic acids (salicylic acid), flavonoids (flavonol heterosides of isorhamnetin and quercetin and isorhamnetin 3-rutinoside), and tannins; terpenoids, including triterpenes (saponins; eg, bidesmosides and monodesmosides of oleanolic acid and phytosterol and taraxasterol), carotenoids (carotenes, lycopenes, violaxanthines, flavoxanthines) and a- and b-amyrins, arnidiol, faradiol, ursadiol, calenduladiol, and heliantriol; lignin). Hydroethanolic extracts prepared by maceration and percolation will contain medium-polar and polar classes such as flavonoids, terpenoid glycosides, carotenoids, coumarins, phenolic acids, and tannins. Supercritical CO2 extracts will contain low-polarity compounds, with some medium polarity ones, but the specific compounds will depend on pressure and co-solvents, such as ethanol. Lipophilic extractions (eg, with vegetable or mineral oil, octyl palmitate) will include lipophilic hydrocarbons; paraffins; fatty acids and fatty acid esters; steroids; tocopherols; apolar carotenoids; mono-, sesqui-, and triterpenoid esters; and triterpene mono-alcohols and diols. The Council of Europe⁷³ Committee of Experts on Cosmetic Products published a monograph on quercetin, noting that this chemical may be found in many plants, including C officinalis. Quercetin, at estimated dietary levels as a dietary supplement (200-1200 mg/d) would not produce adverse health effects. A threshold of toxicologic concern analysis, using conservative assumptions about dermal absorption, applied to a number of chemicals found in C officinalis flowers resulted in systemic exposure values below the threshold of toxicologic concern values established for the relevant Cramer class, except for quercetin, kaempferol, and isorhamnetin. Available NOAELs suggested that these chemicals would not present any toxicity at concentrations used in cosmetics.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	not irritating to slightly irritating
Eye irritation	not irritating to slightly irritating
Skin sensitisation	not sensitising
Acute toxicity	According to CTFA, ³⁴ the median lethal dose (LD50) of C officinalis flower extract for rats was greater than 4.64 g/kg.
Repeated dose studies	Harikumar et al ³⁶ reported a 13-week study of lutein and lutein ester isolated from marigold flowers using male and female Wistar rats. Animals (5 male and 5 female per group) received sunflower oil alone (control—no detectable lutein), or 4, 40, or 400 mg/kg lutein or 4, 40, or 400 mg/kg lutein ester via oral gavage. The authors concluded that lutein and lutein esters are not toxic to rats. Quercetine NOAEL 400 mg/kg bw/day

Mutagenicity/genotoxicity	Ramos et al ⁴⁸ studied the genotoxicity of an extract of <i>C. officinalis</i> L using 3 short-term assays: an Ames test in <i>S. typhimurium</i> TA1535, TA1537, TA98, and TA100, with and without metabolic activation; a mitotic segregation assay in a diploid strain (D-30) of <i>Aspergillus nidulans</i> ; and a mouse micronucleus test. The authors reported that there was no increase in mutation frequency in any of the Ames tests, with or without metabolic activation, for any concentration of the extract. In each case, the positive controls yielded the expected results. A statistically significant and concentration-dependent increase in mitotic segregation (sectors per colony) was reported. The authors also reported a dose-dependent increase in the number of adjacent color sectors that they considered indicative of somatic crossing over. The authors reported no increase in micronuclei in polychromatic cells and no change in the ratio of polychromatic to normochromatic cells.
Carcinogenicity	The International Agency for Research on Cancer (IARC) evaluated the human carcinogenicity potential of coumarin, concluding that this chemical is not classifiable as to its carcinogenicity to humans. ⁴⁹ The evaluation did state that there is limited evidence in experimental animals for the carcinogenicity of coumarin. In particular, in mice of 1 strain, it produced increases in lung adenomas and carcinomas in both males and females and in hepatocellular adenomas in females. There was no increase in tumor incidences in another strain of mouse. In 1 study in rats, coumarin produced a low incidence of renal tubule adenomas in males, seen only after step sectioning of the kidney. Hard et al ⁵⁰ reevaluated kidney tumors and renal histopathology in a 2-year rat carcinogenicity study (National Toxicology Program [NTP]) of quercetin in male and female rats given 0, 1000, 10 000, and 40 000 ppm daily. Quercetin, in this model, exacerbates chronic nephropathy, leading to renal tumors and was considered a secondary mechanism for renal tumor development. The authors also suggested that the single adenocarcinoma, along with a group of 4 other lesions (ranging from hyperplasia to adenomas) had a unique phenotype associated with neoplasms of spontaneous and familial origin. After reviewing available metabolic and other data, including the Hard et al ⁵⁰ review of an NTP 2-year carcinogenicity study presented earlier in this document, the authors concluded that the weight of evidence supports a finding that quercetin, at estimated dietary levels as a dietary supplement (200-1200 mg/d) would not produce adverse health effects.
Reproductive toxicity	28-day subchronic toxicity study using rats. Coriander was given by gavage at 160, 400, and 1000 mg/kg per d. There were reported increases in liver weights, degenerative lesions in the renal cortex, and a high incidence of slight periportal hepatocellular cytoplasmic vacuolization. The no observed effect level (NOEL) was determined to be 160 mg/kg per d. The review concluded that the adverse effects on reproduction and development occurred at maternally toxic levels and not at levels that were not maternally toxic.
Photosensitivity	Not phototoxic
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/21622/4/8 , 2. CIR Expert Panel Final Report of the Cosmetic Ingredient Review Expert Panel Amended Safety Assessment of Calendula officinalis-Derived Cosmetic Ingredients,

	https://online.personalcarecouncil.org/ctfa-static/online/lists/cir-pdfs/PRS514.pdf
POD (NOAEL)	400 mg/kg bw/day, 13 weeks, rat

INCI	CHELIDONIUM MAJUS EXTRACT
Presence in the annexes	None
Description	Chelidonium Majus Extract is an extract of the leaves and aerial parts of the Celandine, <i>Chelidonium majus</i> L., Papaveraceae. Like other members of the Papaveraceae (poppy) family, <i>Chelidonium majus</i> L contains alkaloids as its main active compounds. The alkaloids are regarded as poisonous. The leaves and roots contain 0.2-2.8 % alkaloids. The alkaloids found in greatest quantity is chelidonine, berberine, allocryptopine, coptisine, magnoflorine, sparteine, chelerythrine, sanguinarine and protopine. Use in pharmaceuticals: The usual daily dose is 12 to 30 mg of its active ingredients, which equals to 2-4 g of the extract (Council of Europe, 2008). The European Pharmacopoeia (01/2008:1861) monograph provides the following definition for <i>Chelidonii herba</i> : dried, whole or cut aerial parts of <i>Chelidonium majus</i> L. collected during flowering. Content: minimum 0.6 per cent of total alkaloids, expressed as chelidonine (C ₂₀ H ₁₉ NO ₅ ; Mr 353.4) (dried drug). Therefore the benefit-risk assessment of oral use of <i>Chelidonium majus</i> must be considered negative.
Oral toxicokinetics	In summary, all studies documenting organ distribution as well as faecal excretion of <i>Chelidonium</i> alkaloids show very high faecal and very low hepatic concentrations, thus clearly showing that despite quick hepatic clearance there is no accumulation in the liver. Urinary excretion is consequently low, as has also been shown in humans. This strongly suggests that a very poor absorption, but not the high first pass effect, is mainly responsible for the low systemic bioavailability. This view is also supported by the high sensitivity of the alkaloids sanguinarine, chelerythrine and coptisine to nucleophilic attack, shown e.g. by Debiton et al. (2003) and Ulrichova et al. (2001), which hinder their intestinal absorption. As these publications suggest, metabolic inactivation by conjugation with glutathione also seems to play an important role for the low bioavailability
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	Contact dermatitis has been reported in a woman who had used <i>Chelidonium majus</i> to treat warts. The woman experienced

	severe itching and erythema with papules at the application site. The reaction resolved within a few days without treatment
Eye irritation	N/A
Skin sensitisation	N/A
Acute toxicity	The LD50 in mice was tested and determined as 1300 mg/kg, while in rats it was more than 2000 mg/kg (
Repeated dose studies	Mheddhbi (2002) performed a 4-week toxicity study by oral route (gavage) in rats. The oral toxicity of repeated doses of Chelidonium majus extract (mother tincture according to HAB, containing 0.147% total alkaloids, calculated as chelidonine) over 4 weeks was determined in 20 rats per group (10 males and 10 females) under GLP conditions and according to EU recommendations. The NOEL of Chelidonium fluid extract is the highest administered dose, 1820 mg/kg body weight/day, corresponding to 2.68 mg/kg body weight total alkaloids.
Mutagenicity/genotoxicity	Sanguinarine was shown to induce DNA damage in mouse bone marrow cells (Ansari et al. 2006). In vitro DNA adducts formation with sanguinarine and chelerythrine in the presence of rat hepatic microsomes has been described (Stiborova et al. 2002). Sanguinarine elicited weak positive responses in the Salmonella mutagenicity test after metabolic activation. Genotoxicity in vivo of Macleaya cordata (Willd.) R.Br. (Papaveraceae) was tested. The plant species contains sanguinarine. An isoquinoline mixture made from the plant was given to rats (5,330 or 660 ppm isoquinoline alkaloids), mixed with the food. DNA adducts formation in liver was analysed. No DNA damage was seen to rat lymphocytes or hepatocytes after 90 days administration (
Carcinogenicity	Several in vitro studies show cytotoxicity of several of the individual alkaloids on carcinoma cell lines (Council of Europe, 2008). Ethanol extracts of Chelidonium majus show cytotoxicity on human lymphoblastoid Raji cells, suppressing cell growth at concentrations of 10 and 50 µg/ml (Spiridonov et al., 2005).
Reproductive toxicity	No teratogenicity effects have been seen after use of Chelidonium majus extract (Council of Europe, 2008).
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. EMA Assessment report on Chelidonium majus L., herba, https://www.ema.europa.eu/en/documents/herbal-report/final-assessment-report-chelidonium-majus-l-herba_en.pdf
POD (NOAEL)	NOEL 1820 mg/kg bw/day, 4 weeks, rat

INCI	GALIUM VERUM EXTRACT
Presence in the annexes	None
Description	Due to beneficial effects of Galium species noticed over time, there are now galenic remedies and dietary supplements considered to help in several health disorders, mainly involving immune system, anti-inflammatory processes, detoxication and oxidative stress.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	In a study to assess the skin penetration potential of ethanol, an in vitro study was carried out using excised pig skin and radiolabelled ethanol. Ethanol penetration was greater under occlusion conditions than non-occlusive conditions, as might be expected. Absorption rates were around 21% and 1% of applied doses respectively.
Skin irritation	N/A
Eye irritation	ethanol extracts are irritants
Skin sensitisation	N/A
Acute toxicity	LD50 450mg/kg bw/day
Repeated dose studies	N/A
Mutagenicity/genotoxicity	N/A
Carcinogenicity	N/A
Reproductive toxicity	N/A
Photosensitivity	N/A
Endocrine effects	N/A
Reference	N/A
POD (NOAEL)	30mg/kg bw/day hypothetical

INCI	QUERCUS ROBUR BARK EXTRACT
Presence in the annexes	None
Description	In the European Pharmacopoeia, it is described as the cut and dried bark of young branches and the lateral shoots of Quercus robur. It contains a minimal amount of 3% of tannins, expressed as pyrogallol, calculated with reference to the dried herbal. The oak bark contains highly variable amount of tannins (8-20%). The tannin content depends on the time of the harvest, age of the branches and on the method of assay used. Tannins are polyphenolic secondary metabolites of higher plants. They comprise either: galloyl esters and their derivatives (gallotannins, ellagitannins and complex tannins) or they are oligomeric and polymeric proanthocyanidins and can possess different interflavanyl coupling and substitution patterns. As hydrolysable tannins are considered to present greater risk to health, several cases of acute poisonings of ruminants were noted due to ingestion of fresh leaves of several oak species. The poisoning may be due to hydrolysable tannins and simple phenols. The levels of tannins and condensed tannins were 97.7 mg tannic acid equivalent and 5.8 mg catechin equivalent/g of dry leaves. The ability of tannins to form strong complexes with proteins is considered as their most important activity for both therapeutic and toxicological effects. The available data are sufficient to include the traditional use of specified preparations of oak bark in a Community herbal monograph. Oak bark fulfils the requirement of therapeutic use for at least 30 years (15 years within the European Union, Directive 2004/24/EC). It can be concluded that the benefit/risk assessment for oak bark preparations is positive for use in therapeutical dosages in specific conditions of mild diarrhoea, in minor inflammatory conditions of the

	oral mucosa or skin and in conditions of medically diagnosed haemorrhoids before anorectal use. Tannic acid is authorised in the EU as a food flavouring, although no data on exposure are available for the European population. The Food and Drug Administration estimates that the added tannic acid in the USA can reach 100 mg/kg in various foods
Oral toxicokinetics	The metabolism of polymeric proanthocyanidins by human colonic microflora has been investigated in vitro in anoxic conditions using ¹⁴ C-labelled purified proanthocyanidin (Déprez et al. 2000). Polymers were degraded after 48 h of incubation. As metabolites, low-molecular-weight aromatic compounds: phenylacetic, phenylpropionic and phenylvaleric acids were recognized by gas chromatography. The results show, that proanthocyanidins can be metabolized by the colonic microflora into low-molecularweight aromatic acids. Evaluation of polyphenol bioavailability was tested in vitro in isolated segments of the rat small intestine (Carbonaro et al. 2001). As model compounds, tannic acid and catechin were used. The results indicated a significant, concentration-dependent uptake of both compounds by the small intestinal wall. Higher uptake (50%) was recorded for tannic acid compared with catechin (30%). However, only catechin demonstrated a complete transfer through the gut wall, whereas tannic acid remained in the gut wall via interaction with wall proteins. Both polyphenols were bound significantly by proteins in the intestinal lumen.
Inhalation toxicokinetics	Nasal cancer risk associated with occupational exposure to wood dusts via inhalation cannot be applied to the oral use of oak bark preparations.
Dermal absorption	N/A
Skin irritation	N/A
Eye irritation	Specific IgE to Quercus ilex and Quercus robur and conjunctival challenge test to Quercus robur pollen was positive in two patients
Skin sensitisation	Sensitisation to pollen of Quercus ilex and Quercus robur was found in two patients suffering from rhinoconjunctivitis and seasonal bronchial asthma between April and June
Acute toxicity	Acute toxicity (LD50) of tannic acid given intragastrically to rats was found to be 2.26±0.083 g/kg body weight.
Repeated dose studies	To investigate the response of pancreas to tannins and changes in activities of enzymes in the lumen of the small intestine, broiled cockerels were fed on increasing levels of tannin-containing diet (Ahmed et al. 1991). The birds were fed with a diet consisted mainly of hydrolysable gallotannins for 4 weeks at levels of 13.5, 25 and 50 g/kg. Weight of pancreas was significantly increased with increasing dose of tannins. Activity of trypsin and α -amylase was also dose-dependently increased. Activities of dipeptidase and sucrose α -glucosidase (disaccharidase) in the intestinal mucosa were inhibited. Growth of the birds fed on tannins diet was adversely affected. In two studies with New Zealand male rabbits, inclusion of a tannic acid preparation (from chestnut wood) for 21 days at 5 000 mg or 10 000 mg/kg feed showed no adverse effects on growth or feed conversion, which were significantly better than the control group that was not given tannic acid. The published studies show that levels of tannic acid preparations up to 15 000 mg/kg feed for adult ruminants and 10 000 mg/kg feed for laying hens do not result in adverse effects in these species. Rabbits tolerate up to 10 000 mg/kg, pigs for fattening tolerate up to 1 500 mg/kg and chickens for fattening tolerate up to 1 000 mg/kg without adverse effects. Young animals seem to be more sensitive (> 1 500 mg/kg feed was harmful to calves). In the study with weaned piglets, animals receiving 125 mg/kg showed no sign of adverse effects. In all cases, the values shown to be safe are several orders of magnitude higher than the proposed use as feed flavouring.
Mutagenicity/genotoxicity	Oak tannins are genotoxic in cultured human embryonic MRC-5 lung cells (Zhou et al. 1995). Authors evaluated the ability of solvent extracts of natural

	woods to induce chromosome aberrations. In experiments three concentrations per extract with or without metabolic activation and 100 metaphase cells were examined. No dose dependent activity was found with tested extracts in the presence of S9, but dose-dependent chromosomal and chromatid breaks were caused by oak wood. No metabolic activation was required for their effect. Labieniec and Gabryelak (2003) showed that tannic, ellagic and gallic acids have genotoxic and cytotoxic properties in Chinese hamster cell line B14 when using the Comet assay for detection of DNA damage. Tannic, ellagic and gallic acids were exposed at concentration range of 15 – 240 µM. Tannins decreased viability of the cells, with highest cytotoxicity at the concentration of 60 µM. The formation of DNA single-strand breaks was observed.
Carcinogenicity	Nasal cancer risk associated with occupational exposure to wood dusts via inhalation cannot be applied to the oral use of oak bark preparations.
Reproductive toxicity	N/A
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. EMA Assessment report on Quercus robur L., Quercus petraea (Matt.) Liebl., Quercus pubescens Willd., cortex 2. EFSA Scientific Opinion on the safety and efficacy of tannic acid when used as feed flavouring for all animal species
POD (NOAEL)	Hypothetical value of 100mg/kg bw/day as worst case from FDA

INCI	SALVIA OFFICINALIS LEAF EXTRACT
Presence in the annexes	None
Description	According to the EMA monograph, the principal components of Salvia officinalis oil are thujone, 1,8-cineole, and camphor. The main classes of constituents in Salvia officinalis oil are identified as terpenoids, hydroxycinnamic acid derivatives, flavonoids, phenolic glycosides, and polysaccharides. According to the 49th Amendment of the International Fragrance Association (IFRA) standards, in the absence of analytical data, thujone is expected to occur naturally from 2-10% in Salvia officinalis oleoresin and from 8 – 33% in Salvia officinalis oil. CIR Expert Panel noted the GRAS status and historical food uses of Salvia officinalis (sage)-derived ingredients, especially Salvia officinalis leaves, and agreed that systemic exposures from food would be much higher than those from cosmetic use.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	irritating
Eye irritation	irritating
Skin sensitisation	sensitising
Acute toxicity	LD 50> 2000 mg/kg bw
Repeated dose studies	In an 8-wk study using groups of 5 white rats (sex and strain not specified), a daily dose of 250 mg/kg bw Salvia officinalis oil was well tolerated when given by oral administration. Upon increase of the daily dose to 500 mg/kg bw/d, convulsions occurred in some animals. Upon increase to 1000 mg/kg bw/d, most animals died, and all animals died when the dose was increased to 1250 mg/kg bw/d (timing and duration of all 3 dose increases not provided). The

	no-observed adverse effect level (NOAEL) of Salvia officinalis oil was determined to be 250 mg/kg bw/d, under the conditions of this study.
Mutagenicity/genotoxicity	Ames test: non mutagenic
Carcinogenicity	N/A
Reproductive toxicity	Twenty-four female ICR mice received a daily dose of 0.25% Salvia officinalis oil (equivalent to 375 mg/kg/d), in rodent feed, for 14 d. A significant decrease in embryo cell distribution, according to nucleus number, was observed in dams which consumed Salvia officinalis oil (until day 4 of gestation), compared to control females fed with 1% edible soya oil. Dam weights and the proportion of dead cells in embryos were not affected. Groups of 7 female Wistar rats were administered either distilled water or 30 mg/kg bw/d hydroalcoholic extract (70% ethanol) of Salvia officinalis, orally, for 30 d. No significant differences in blood estradiol or progesterone were observed, and a decrease in the duration of estrous cycles in Salvia officinalis extract-treated rats was not statistically significant. An increased number of alveolar buds and lobules in the whole mount slides, as well as an increase in the number and diameter of ducts in the histological sections of rats treated with Salvia officinalis extract, were statistically significant.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/25026/7/7/1 2. CIR Expert Panel Safety Assessment of Salvia officinalis (Sage)-Derived Ingredients as Used in Cosmetics
POD (NOAEL)	hypothetical 20mg/kg bw/day from 30 days reprod.toxic as worst case

INCI	EUCALYPTUS GLOBULUS LEAF OIL
Presence in the annexes	None
Description	Eucalyptus globulus oil is an UVCB substance, mostly composed of 1,8-cineol, (d,l)-limonene and alpha-pinene.
Oral toxicokinetics	The physical chemical characteristics described above suggest that the three major constituents of Eucalyptus globulus oil are of adequate molecular size to participate in endogenous absorption mechanisms within the mammalian gastrointestinal tract. Due to its low molecular weight, moderate lipophilicity and high water solubility, 1,8-cineol may be absorbed in the gastro-intestinal tract through aqueous pores or carriage across membranes with the bulk passage of water.
Inhalation toxicokinetics	1,8-cineol is not volatile. Physico-chemical properties of alpha-pinene indicate that it may be available as a vapour can be taken up by micellar solubilisation and reach the lower respiratory tract.
Dermal absorption	Dermal absorption of 1,8-cineol is expected to be significant based its log Pow and water solubility. (d,l)-limonene and alpha-pinene being more lipophilic (log Kow = 4.5), their rate of uptake into the stratum corneum is expected to be high while their rates of penetration is likely to be limited by their rate of transfer between the stratum corneum and the epidermis. It is assumed that the dermal uptake is also limited by their slight water solubility. However, (d,l)-limonene and alpha-pinene being both identified as skin sensitiser, some uptake must occur although it may only be a small fraction of the applied dose. Enhanced penetration might also be expected since both substances are skin irritants. Beta-pinene absorption into the different skin layers is rapid (steady-state concentrations in the skin obtained after 1-h exposure) but do

	not permeate through the skin to the acceptor medium due to large accumulation into the skin tissue.
Skin irritation	irritating
Eye irritation	not irritating
Skin sensitisation	May cause sensitisation by skin contact
Acute toxicity	LD50 = 3320 mg/kg bw
Repeated dose studies	In an OECD 422 study, NOAELs for systemic toxicity were considered to be 300 and 1000 mg/kg bw/day for females and males rats, respectively.
Mutagenicity/genotoxicity	Ames test: non mutagenic (OECD 471, GLP, K, rel. 1)
Carcinogenicity	N/A
Reproductive toxicity	In an OECD 422 study, NOAELs for systemic toxicity were considered to be 300 and 1000 mg/kg bw/day for females and males rats, respectively.
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/14864/7/9/1
POD (NOAEL)	300mg/kg bw/day from 2 weeks repeated dose study

INCI	MENTHA PIPERITA OIL
Presence in the annexes	0
Description	Mentha Piperita (Peppermint) Oil has been extracted (distilled; water solvent) from the leaves of Mentha piperita harvested (first in July and second harvest in September). The major constituents of Mentha Piperita (Peppermint) Oil include the terpenes: (-)-menthol (30 - 55%), (-)-menthone (14 - 32%), (+)-isomenthone (1.5 - 10%), (-)-menthyl acetate (2.8-10%), (+)-menthofuran (1.0 - 9.0%), and 1,8-cineol (3.5 - 14%). A public statement from the European Medicines Agency on the use of herbal medicinal products containing pulegone and menthofuran indicated that Mentha Piperita (Peppermint) Oil contains a maximum of 4% pulegone and between 1% and 9% menthofuran. In the European Union, the highest recommended daily dose of Mentha Piperita (Peppermint) Oil is 1.2 ml, i.e., 1080 mg Mentha Piperita (Peppermint) Oil (contains a maximum of 140 mg pulegone + menthofuran).¹⁶ For a 60 kg person, this would correspond to a daily intake of 2.3 mg/kg body weight. This recommended daily dose of Mentha Piperita (Peppermint) Oil in medicinal products results in pulegone/menthofuran that exceeds the tolerated daily intake (TDI) (0.1 mg/kg) that was established for food by the Committee of Experts on Flavoring Substances (CEFS). A maximum dermal use level of 5.4% has been recommended for Mentha Piperita (Peppermint) Oil.⁷⁶ This dermal restriction is based on 8% menthofuran (pulegone metabolite) and 3% pulegone content, with limits of 0.5% for menthofuran and of 1.2% for pulegone. The authors also recommended that Mentha Piperita (Peppermint) Oil, due to menthol content, should be avoided altogether in cases of cardiac fibrillation and in individuals with a glucose-6-phosphate dehydrogenase deficiency.
Oral toxicokinetics	Mentha Piperita (Peppermint) Oil is relatively rapidly absorbed after oral administration and eliminated mainly via the bile. Based on physical properties of the constituents, absorption via the oral, inhalation and dermal

	route is expected to be moderate to high. For risk assessment, a default absorption of 100% is assumed.
Inhalation toxicokinetics	Based on physical properties of the constituents, absorption via the oral, inhalation and dermal route is expected to be moderate to high. For risk assessment, a default absorption of 100% is assumed.
Dermal absorption	Based on physical properties of the constituents, absorption via the oral, inhalation and dermal route is expected to be moderate to high. For risk assessment, a default absorption of 100% is assumed.
Skin irritation	Hairless sites on 5 white rabbits were injected intradermally with 0.05 ml Mentha Piperita (Peppermint) Oil. Gross examinations were performed at 24 h and 48 h, at 1 and 2 weeks, and, in some cases, at 1 month after dosing. Dosing was repeated between 5 and 10 times. At microscopic examination of skin samples, moderate reactions characterized by polymorphonuclear leucocytes, lymphocytes, and plasma cells (without necrosis) were observed in 3 rabbits. Severe reactions, which were marked by the above as well as necrosis, were observed in the other 2 rabbits. Positive patch test reactions to Mentha Piperita (Peppermint) Oil and Mentha Piperita were reported in 8 of 9 case reports on patients with diseased skin
Eye irritation	In the in vitro Bovine Corneal Opacity and Permeability test, cornmint oil (mentha arvensis) did not induce ocular irritation through both endpoints, opacity and permeability, resulting in a mean in vitro irritancy score of 4.3 after 10 minutes of treatment.
Skin sensitisation	Several studies on the skin sensitising potential of Cornmint oil were available for the dossier, which are read across to peppermint oil. Because of the comparable reliability between the studies and the differing outcomes, a Weight of Evidence approach was used to describe the endpoint of skin sensitisation. Three of the four animal studies indicated sensitisation and the human study indicated skin sensitisation in two subjects.
Acute toxicity	In a study involving fasted mice, an LD50 of 2410 mg/kg was reported for Mentha Piperita (Peppermint) Oil diluted in olive oil.
Repeated dose studies	Groups of 28 Wistar rats were given oral doses of 10, 40, and 100 mg/kg Mentha Piperita (Peppermint) Oil (diluted with soybean oil) daily for 90 days. The no-observed-adverse-effect level (NOAEL) for Mentha Piperita (Peppermint) Oil was 40 mg/kg/day in this study.
Mutagenicity/genotoxicity	he mutagenic potential of Mentha Piperita (Peppermint) Oil was investigated using the Salmonella/mammalian microsome test.1 The following Salmonella typhimurium strains were used: TA1535, TA100, TA1537, and TA98. The sample tested contained 38.1% menthol, 33.7% menthone, and 1.7% pulegone; the remaining components were not identified. Mentha Piperita (Peppermint) Oil, tested at doses of 6.4, 32, and 160 µg/plate, produced the same number of revertants as the negative control. Toxicity was noted at the next (and maximum) dose of 800 µg/plate. Metabolic activation appeared to have made the oil less toxic to the bacteria. Mentha Piperita (Peppermint) Oil was not genotoxic.
Carcinogenicity	In a carcinogenicity study of toothpaste and its components, groups of 52 male pathogen-free CFP (IClredefined) mice were dosed by gavage with 4 or 16 mg Mentha Piperita (Peppermint) Oil/kg/day, 6 days per week for 80 weeks. Treatment was followed by a 16- to 24-week observation period. An untreated group of 52 male mice and a vehicle control group of 260 male mice that received the toothpaste base (which did not contain chloroform, eucalyptol, or Mentha Piperita (Peppermint) Oil) were maintained as controls. At least one neoplasm at any site was observed in 73%, 69%, 65%, and 71% of mice of the low-dose, high-dose, untreated control, and vehicle control groups, respectively. The incidence of neoplasms of the lungs and kidneys was comparable among mice of the treated and nontreated groups. Hepatic cell

	tumor incidence for Mentha Piperita (Peppermint) Oil-dosed mice (25%) was comparable to the incidence for mice of the vehicle control group (27%); the incidence for the untreated group was 19%. Malignant lymphoma was found in 25%, 21%, 10%, and 14% of mice of the low-dose, high-dose, untreated, and vehicle control groups, respectively. The researchers did not discuss whether the differences in tumor incidence were significant.
Reproductive toxicity	Based on the absence of effects on development and reproduction in the major and some minor constituent, cornmint oil and peppermint oil are not expected to be toxic to the development or reproduction.
Photosensitivity	Undiluted Mentha Piperita (Peppermint) Oil was applied to the backs of 6 Skh:hairless mice. Thirty minutes later, the mice were irradiated for either 1 h with light from a fluorescent blacklight at an integrated UVA of 3 W/m ² , or for 40 minutes with light from a Xenon lamp at a weighted erythema energy of 0.1667 W/m ² . The mice were examined at 4 h, 24 h, 48 h, 72 h, and 96 h after radiation treatment. No effects were noted
Endocrine effects	N/A
Reference	1. CIR Expert Panel Amended Safety Assessment of Mentha piperita (Peppermint)-Derived Ingredients as Used in Cosmetics, https://online.personalcarecouncil.org/ctfa-static/online/lists/cir-pdfs/FR741.pdf 2. ECHA https://echa.europa.eu/registration-dossier/-/registered-dossier/14548/7/9/1
POD (NOAEL)	40mg/kg bw/day from 90 days oral study on rats

INCI	THYMUS VULGARIS FLOWER/LEAF OIL
Presence in the annexes	None
Description	Thymus vulgaris L. is widely used as an ingredient in cooking and in herbal medicine.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	Irritant
Eye irritation	Irritant
Skin sensitisation	Sensitizer
Acute toxicity	The LD50 of the essential oil oral route is 2.84 g/kg body weight in rats
Repeated dose studies	The results of the 28-day oral toxicity test suggest that the no-observed-adverse effect level (NOAEL) is greater than 250 mg/kg/day.
Mutagenicity/genotoxicity	Thyme essential oil had no mutagenic or DNA-damaging activity in either the Ames test (strains TA1535, TA1537, TA98, TA100, with and without metabolic activation, 0.25 and 0.5 µl essential oil/plate) or Bacillus subtilis rec-Assay
Carcinogenicity	N/A
Reproductive toxicity	N/A
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. EMA, Assessment report on Thymus vulgaris L., vulgaris zygis L., herba
POD (NOAEL)	250mg/kg bw/day, oral, rat, 28 days

INCI	ILLICIUM VERUM LEAF OIL
Presence in the annexes	None

Description	Several publications retrieved by the applicant consistently reported the occurrence of estragole (2–6%) in essential oils from the fruit of <i>I. verum</i> . Two publications also reported the presence of methyleugenol (0.12%) and safrole (0.21%) in essential oils from star anise fruit obtained by different manufacturing process and containing, respectively, 10% and 14.4% estragole. Many of the components of star anise oil, accounting for about 95% of the % GC peak areas, have been previously assessed and considered safe for use as flavourings, and are currently authorised for use in food 9 without limitations and for use in feed 8 at individual use levels higher than those resulting from the intended use of the essential oil in feed.
Oral toxicokinetics	N/A
Inhalation toxicokinetics	N/A
Dermal absorption	N/A
Skin irritation	irritant
Eye irritation	irritant
Skin sensitisation	sensitizer
Acute toxicity	0
Repeated dose studies	The FEEDAP Panel identified a no observed adverse effect level (NOAEL) of 10 mg/kg bw per day for non-neoplastic lesions (changes in organ weight 29 and function, including effects on liver and the glandular stomach) from a 90-day study in mice with methyleugenol
Mutagenicity/genotoxicity	N/A
Carcinogenicity	N/A
Reproductive toxicity	N/A
Photosensitivity	N/A
Endocrine effects	N/A
Reference	1. EFSA Safety and efficacy of a feed additive consisting of an essential oil derived from the fruit of <i>Illicium verum</i> Hook.f. (star anise oil) for use in all animal species (FEFANA asbl)
POD (NOAEL)	10mg/kg bw/day

9) Undesirable effects and serious undesirable effects

The responsible person is informed about the obligation to collect information on undesirable effects and serious undesirable effects, as well as to enter them in the Cosmetic product safety report.

10) Information on the cosmetic product

Cosmetic product DERMOTIN BLOWART contains ingredients of natural origin in which, according to data from the existing scientific literature, the presence of allergens is possible and is indicated as a challenge in assessing the safety of the products that contain them.

The product DERMOTIN BLOWART was tested for allergens presence, the results of which are listed in the attachment to this document.

The chemical substances and mixtures used to produce cosmetic product DERMOTIN BLOWART are widely used as ingredients for cosmetic products, they are used in concentrations that are acceptable from the point of view of safety assessment, based on which it can be concluded that the product will not cause irritation and sensitization of the skin. Eye irritation may occur in case of contact with the eyes. The product packaging clearly states the warning measure to avoid contact with the eyes.

Cosmetic product DERMOTIN BLOWART doesn't contain:

- Substances and mixtures that have been tested on animals.
- Intentionally added CMR substances
- Prohibited substances listed in the annexes of the Regulation (EC) 1223/2009
- Intentionally added nanoparticles

Part B. Cosmetic Product Safety Assessment

1) Assessment conclusion

The cosmetic product DERMOTIN BLOWART is safe for human health when used under normal or reasonably foreseeable conditions, in accordance with the Cosmetic Regulation (EC) 1223/2009.

2) Labelled warnings and instructions of use

Cosmetic product DERMOTIN BLOWART is intended to be used as skin care cream without subsequent washing after application.

Instructions for use written on the cosmetic product:

„Apply the cream to anal area 1-2 times daily.“

Warnings and precautions written on the cosmetic product:

„Sensitivity to any ingredient. Avoid contact with eyes. If the contact occurs, wash with plenty water. In case the irritation does stop ask for a doctor's advice“.

The cosmetic product DERMOTIN BLOWART is presented in a way that clearly communicates its purpose. Precise and clearly understandable instructions are given, based on which it is considered that it cannot be misused.

The cosmetic product DERMOTIN BLOWART has characteristics in terms of appearance, odor, packaging design, appearance and content of the labelling and volume, and other characteristics of general appearance that cannot mislead the consumer about the purpose of the product and the way of its use.

3) Reasoning

Safety assessment of the cosmetic product DERMOTIN BLOWART was done considering all available relevant data, up to date regulatory requirements and the latest published scientific knowledge. Facts relevant for the safety assessment:

- To successfully assess the safety of the product, information on the body part and body surface of product application, the amount of product applied, duration and frequency of application, normal and reasonably applicable routes of exposure to the product and substances, and the target population were considered.
- All technical documentation that defines the substances and mixtures used to produce the cosmetic product, as well as the technical documentation that defines the packaging used for the packaging of the cosmetic product, is considered adequate for evaluating the product's safety.
- All information regarding the properties of the cosmetic product, its production method and the results of the conducted experimental tests submitted by the responsible person are considered adequate for evaluating the product's safety. Any further change of any parameter relevant to the product safety assessment must be considered by the responsible person and a revision of this document carried out, of which the responsible person has been informed.
- The toxicological profiles of the substances included in the composition of the cosmetic product were generated considering the latest scientific knowledge.
- All substances for which there is a legal concentration limit, and which are listed in relevant annexes of the Regulation (EC) 1223/2009 meet the legal requirements and for them the Margin of Safety (MoS) was not calculated.
- For substances for which there is no legal concentration limit and there are available data on conducted repeated dose studies, based on which it is possible to obtain the value of the No Observed Adverse Effects Level are expected (NOAEL), the Margin of Safety (MoS) was calculated, which for all given substances is greater than 100 which is considered acceptable.
- For substances for which there is no legal concentration limit and there is no data on repeated dose studies, based on which it is possible to obtain a NOAEL value, an extensive explanation is provided.

- Based on the data on the potential of ingredients substances to cause skin irritation and sensitization, it was concluded that the product DERMOTIN BLOWART is not expected to cause skin irritation and sensitization. It is possible to expect that the product in case of contact with the mucous membrane of the eye will lead to slight eye irritation, which is why it is clearly indicated on the product packaging that contact with the eyes should be avoided.
- There are no expected interactions between the ingredients of the product, as well as interactions between the ingredients of the product and the packaging for the period of 6 months after the production, which would lead to negative outcomes regarding the stability and efficiency of the product, which was confirmed by stability and compatibility studies.
- The method of production and packaging of the cosmetic product excludes the possibility of occurrence of microbiological contamination that would lead to endangering human health from the use of the cosmetic product, which was confirmed by the conducted stability and compatibility studies.
- Cosmetic product doesn't contain prohibited substances listed in the Annex II of the Regulation (EC) 1223/2009.
- Based on the overall appearance of the product and the content listed on the packaging, the purpose of the product as anal region skin care cream is clear. Instructions for use and warning measures are clearly indicated on the packaging.

4) Assessor's credentials and approval of part B

Safety Assessor: Vesna Milanković, FVRM Consulting, Koste Jovanovića 21, 11000 Belgrade, Serbia

Academic and expert qualifications:

- M.Sc. Chemical Engineering, 300 ECTS (Faculty of Technology, University of Novi Sad)
- Specialist Academic Studies, Cosmetology, 60 ECTS (Faculty of Pharmacy, University of Belgrade)
- Safety Assessment of Cosmetics in the EU, 6 ECTS (Vrije University Brussel)
- Chemical Advisor License, Belgrade.

The conclusion of the safety assessment given in part B of this document for the product DERMOTIN BLOWART is correct and the explanation provided is adequate. The assessment was done based on the documentation submitted by the responsible person, up to the current date of updated regulatory requirements and up to the current date of published scientific findings.

Signature:

A handwritten signature in blue ink, appearing to read 'Vesna Milankovic', is written over a horizontal line.

Vesna Milanković